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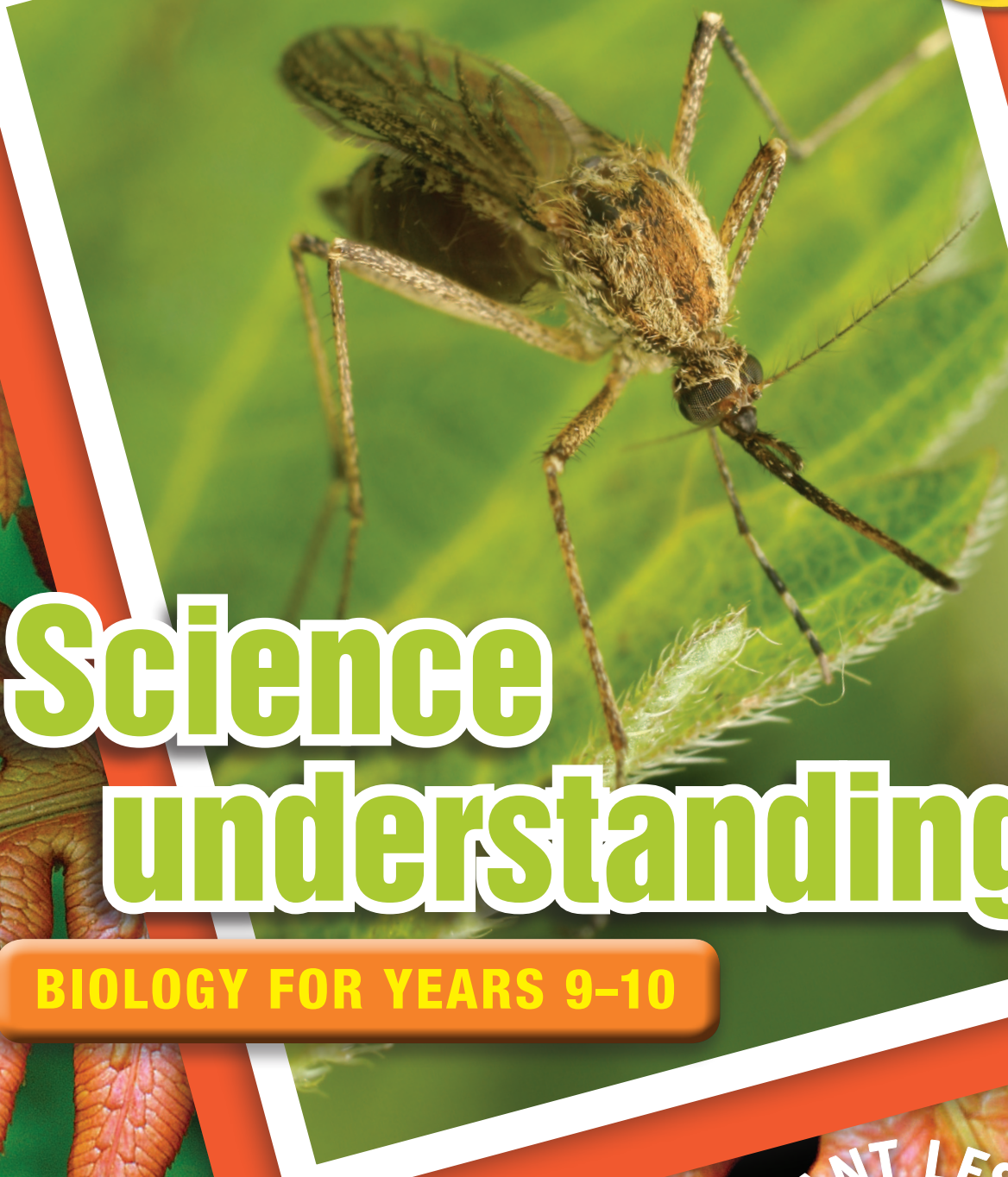

Science understanding

BIOLOGY FOR YEARS 9–10

Beth Montgomery

INSTANT LESSONS

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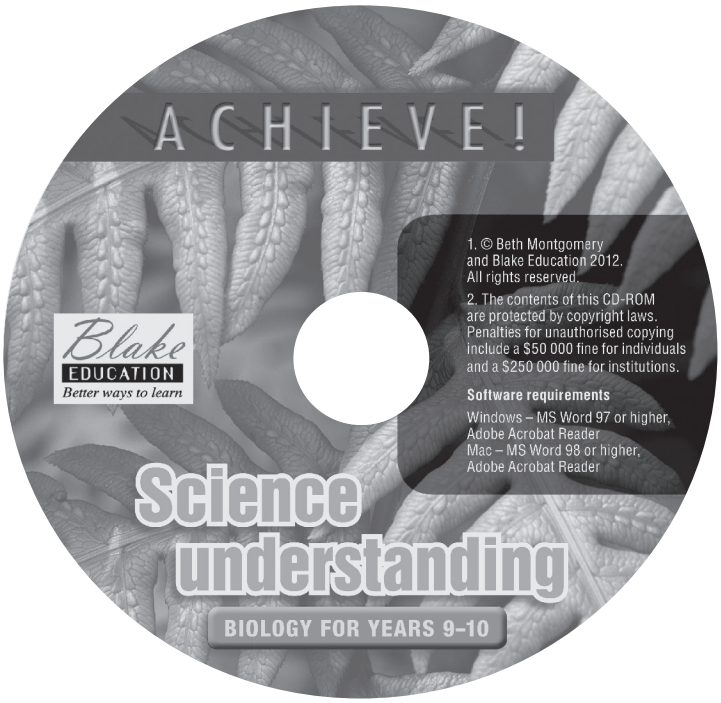
Made for the Australian Curriculum

Achieve! Science

Get ready for the Australian Curriculum with editable work sheets that have been written for the needs of low literacy students.



Features

- ✓ Refreshing strategies for teaching *Science understanding, Biological sciences* from the Australian Curriculum.
- ✓ Activities designed to reinforce and enhance literacy skills.
- ✓ Emphasis on understanding essential science content and concepts.
- ✓ Comprehensive Teacher's Notes that provide clear objectives, background information, starter and extension activities.
- ✓ CD-ROM containing all pages from the book making them ideal for use on Interactive Whiteboards.

Software requirements

Windows *– MS Word 97 or higher, Adobe Acrobat Reader*

Mac *– MS Word 98 or higher, Adobe Acrobat Reader*

- ✓ Word files on the CD-ROM that can be edited to suit the specific needs of students.
- ✓ All illustrations are included on the CD-ROM and can be used on new teacher-created work sheets.

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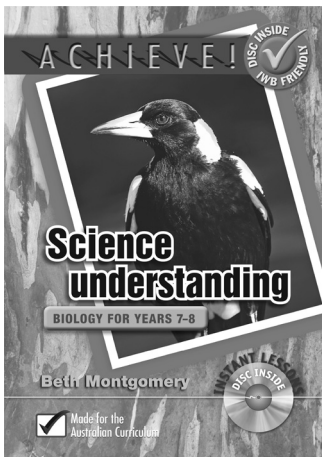

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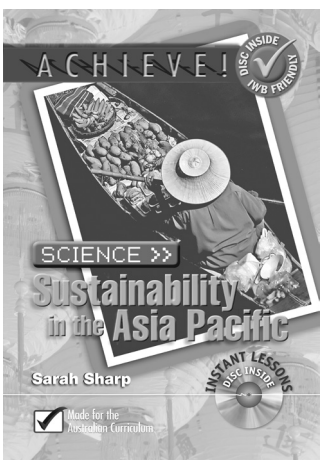

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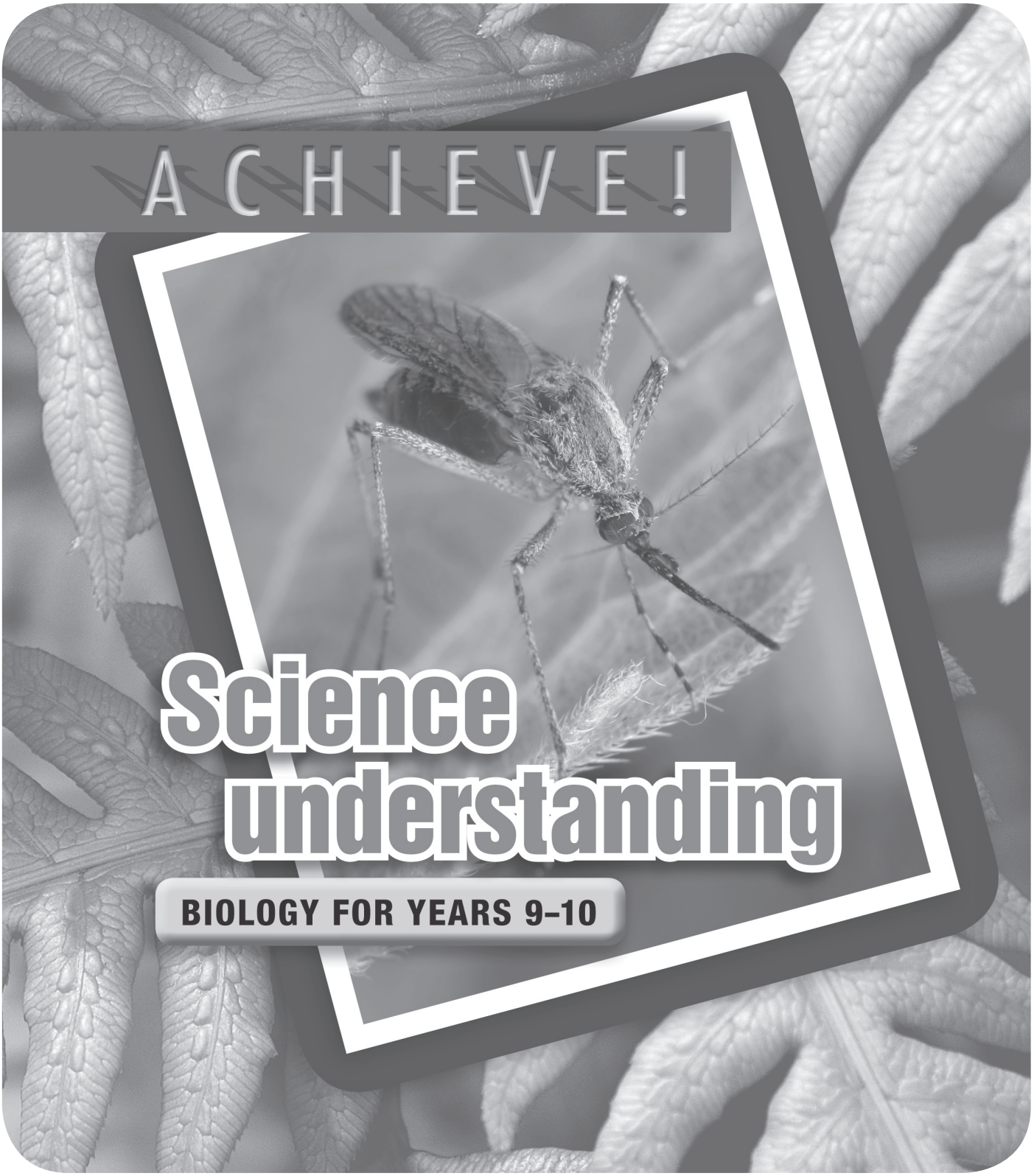

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Introduction

Achieve!

The *Achieve!* series of Science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

This title will assist teachers to develop lessons for the biology sub-strand in *Science understanding* for years 9 and 10. Units of work cover a variety of topics including the impact of diseases like Ross River virus, how deficiencies such as Vitamin D affect different individuals, vaccinations, models for measuring population growth, mitosis and meiosis, and geological time and the fossil record. Work sheets help students build their scientific language knowledge and understanding of key concepts.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the individual needs of students.

This book is divided into ten units. The units can be taught as a lesson with students in groups of varying sizes. Alternatively, a single Activity sheet can be used as support material, for individual homework, assessment or revision.

The teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the work sheets.

Prior knowledge – This refers to the skills and knowledge required for students to complete the tasks. Some activities are more challenging than others, however, as a rule, students should have a reading comprehension age of six to nine years to be able to attempt the activities.

Background – This section provides additional guidance to help teachers to present a successful lesson. It may include definitions or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity and Resource work sheets – The Resource sheets do not always contain activities and are used either to stimulate discussion or as a part of the unit's activities. The Activity work sheets may be used sequentially or as stand-alone tasks, depending on the needs of the student.

Extension ideas – Additional tasks are suggested to recap on the main skills or knowledge taught in the lesson, as well as ways to extend, assess or include technology in the lesson.

The Australian Curriculum correlations

	Year 9	Units									
	Year 10	1	2	3	4	5	6	7	8	9	10
Science understanding **Biological sciences**											
Multi-cellular organisms rely on coordinated and interdependent internal systems to respond to changes to their environment	ACSSU175	✓	✓	✓	✓						
Ecosystems consist of communities of interdependent organisms and abiotic components of the environment; matter and energy flow through these systems	ACSSU176			✓		✓					
The transmission of heritable characteristics from one generation to the next involves DNA and genes	ACSSU184						✓	✓			
The theory of evolution by natural selection explains the diversity of living things and is supported by a range of scientific evidence	ACSSU185								✓	✓	✓

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Teacher's notes – Responding to stimuli

Achieve!

Objectives

- To understand the importance of the brain and nervous system in responding to our environment
- To learn of the history of research into the nervous system
- To identify the relationship between stimulus and response
- To understand that plants are able to respond to stimuli
- To recognise parts of the excretory system

Prior knowledge

Students should have a basic understanding of humans' senses, and their various body systems.

Background

Students must recognise that organisms need to respond to both internal (e.g. full bladder) and external (e.g. hot fire) stimuli. The nervous system coordinates such responses. Stimuli can be to do with pain, temperature, light and sensory input.

Starter activity

Ask students how they know when to go to the toilet, or when to put a coat on? What are the triggers for these events? How does their body react when they touch something hot? Why do some flowers follow the path of the sun, or drop their leaves in autumn? Discuss the terms 'stimulus' and 'response'.

Activity sheets

The Activity sheet 'The human nervous system' explores what our nerves do. Hold a class discussion on this topic. Talk about our senses, how cold or hot we are and how our muscles move. Students have a cloze exercise to complete and a diagram of the human nervous system to label.

The 'Squid reactions' Activity sheet demonstrates that much of our knowledge of how our nerves work comes from scientists examining the large nerves found in squid. Students order sentences explaining this research before answering some true/false questions.

'Pain and pain control' explains that pain is necessary to help the body protect itself. Students are asked to recognise examples of different types of pain by filling in the missing vowels. They then label a diagram to consolidate their learning.

The Activity sheet 'Body temperature' looks at the way many animals are able to live in sub-zero temperatures. They conserve body heat with a variety of adaptations including layers of fat and warm fur. Warm-blooded animals also need to cool down when they are hot. Students have a cloze exercise to complete and a table to fill out. There are also phrases to match.

The 'Human kidneys and excretion' Activity sheet presents a diagram showing parts of the excretory system for students to label. Students then complete a crossword to familiarise themselves with words associated with the kidneys and human excretory system.

In 'Plant responses to temperature', students read a passage highlighting how plants respond to temperature changes. Students underline examples of plants responding to temperature changes. They then illustrate a timeline showing a year in a deciduous tree's life and list examples of deciduous trees.

Extension ideas

Ask students to draw a flow chart poster depicting the relationship between stimulus and response. Direct them to use a specific example to illustrate their flow chart.

Selected answers

Human kidneys and excretion Across: **4** bladder; **6** artery; **8** urine; **9** vein. Down: **1** kidney; **2** urea; **3** glomerulus; **5** urethra; **7** ureter.

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The human nervous system

☞ **1** Use the word bank to complete the passage below:

Word bank

different spinal Nerves back ears backbone sensitive cord changes brain sends muscles central balance messages receive

All **mammals** have a b_____. Inside the backbone runs a long mass of tissue known as the s_____ cord. The spinal c_____ runs from the b_____ inside the skull right down to the tailbone. The brain and **spinal cord** are known as the c_____ nervous system. N_____ run from the spinal cord to every part of the body. Many of these nerves r_____ information from our surroundings. This information is called **stimulus**. The information is sent b_____ to the brain for processing.

There are many d_____ types of nerves. Some nerves are s_____ to touch, some to pain, some to c_____ in temperature. The **optic** nerve s_____ messages to the brain about what we see; nerves in the e____ send messages about the sounds we hear, and also about b_____. The brain responds to these m_____ by sending instructions through our nerves to make our m_____ move or to release **hormones** into our bloodstream.

2 Underline examples of stimulus and circle the examples of how our bodies respond in the text above.

3 On the diagram below, draw and label the parts of the central nervous system described in the passage above.

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Squid reactions

☞ **1** Order the following sentences about scientists studying the central nervous system of the squid, then rewrite the paragraph below.

Thanks to the humble squid, scientists since the 1930s have been able to study in detail how nerves work.

It is about 1 mm in diameter and almost a metre long.

These experiments earned scientists a Nobel Prize in 1963 and form the basis of our understanding of nerve cells today.

Because the nerve is so long, scientists could place **electrodes** inside it to measure changes in current flow across the cell membranes when a nerve fired.

Human nerves are very thin and hard to see, but the squid has the longest nerve of any known creature.

Squid are invertebrates that have a highly developed **sensory** and nervous system.

This nerve is called the giant axon and it is involved in **propelling** the squid through the water quickly.

_____

_____

_____

_____

_____

_____

_____

2 Write whether the following statements are true or false.

a) The giant axon is the largest known nerve. _____

b) The axon is involved in squid vision. _____

c) Human nerves are so thin they are hard to see. _____

d) An electric current could be measured in the axon. _____

e) Humans have a nerve called the giant axon. _____

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Pain and pain control

Pain receptors play a major role in protecting our bodies. When a receptor feels pain, the body takes steps to minimise the pain and protect itself from further damage.

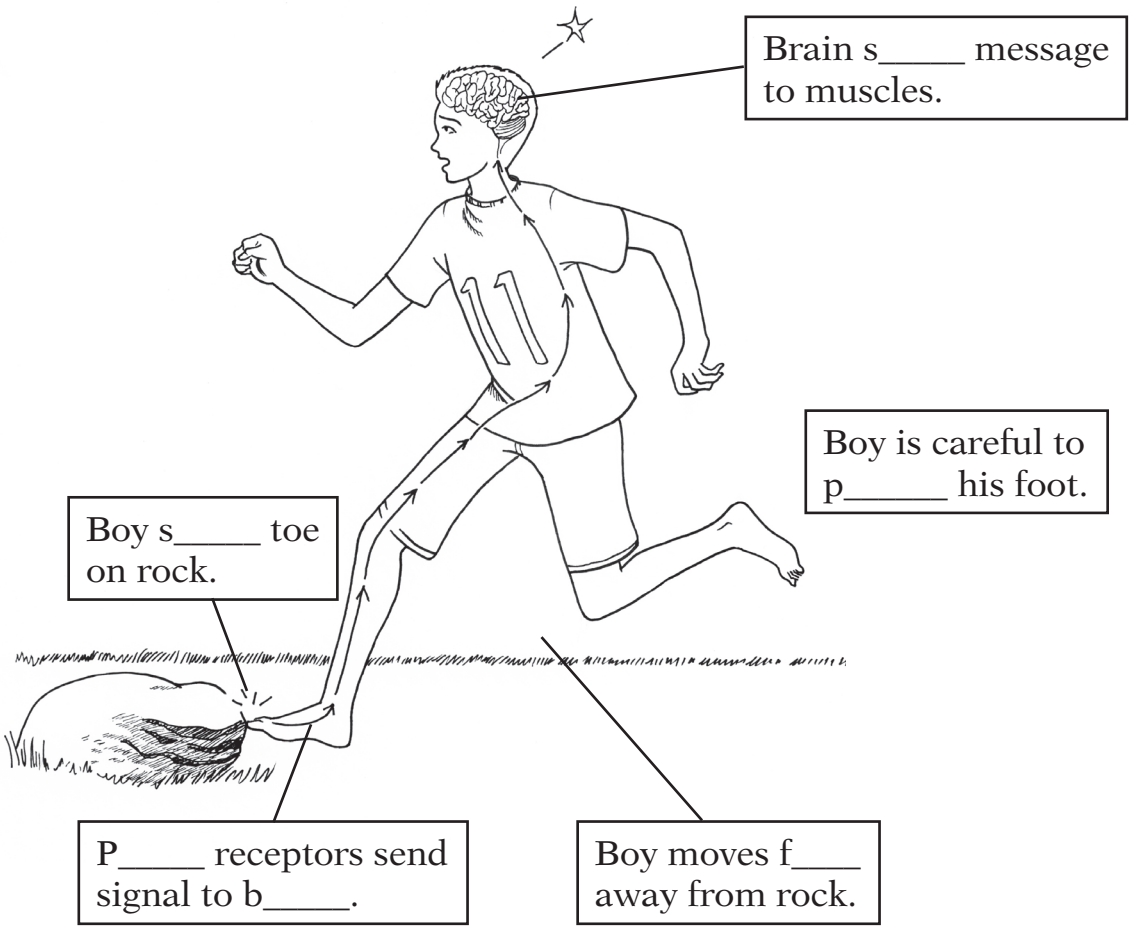
☞ **1** Add the missing vowels (a, e, i, o, u), then match each pain type to its example.

A **response** to hot or c__ld	crush fingers in car door
Ch__m__c__l response	amputees feel pain in missing limb
Mechanical r__sp__ns__	burn fingers on stove
Nerve d__m__g__/**inflammation**	put medicine on a cut and it stings
Phantom p__ __n	sore back or arthritic joints

2 Use the word bank to complete the labels on the diagram below.

Word bank

protect foot brain stubs Pain sends



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Body temperature

☞ **1** Use the word bank to complete the passage below.

Birds and mammals are warm-b_____ animals. They keep their body temperature c_____ despite temperature c_____ in their environment, otherwise they can die. Animals like polar bears and p_____, who live in extreme cold, have large f____ deposits beneath their s_____. The polar bear's thick fur t_____ layers of air to keep it w_____. Both creatures can walk on ice because the blood vessels in their feet are c_____ to prevent heat loss. When humans are very h____ they sweat and their blood v_____ dilate to cool them down.

Word bank

penguins
constant
hot
skin
changes
blooded
fat
traps
warm
constricted
vessels

2 Our body maintains a constant temperature in many different ways. Decide which of the following methods retain heat, and which help the body lose heat, then list them in the table.

shivering sweating fat layer hair covering
dilated blood vessels **constricted** blood vessels

How the body retains heat	How the body loses heat

3 Match the phrases below.

Penguins have large fat	to keep the body warm.
Shivering generates heat	the body in hot weather.
Sweating cools	deposits under their skin.

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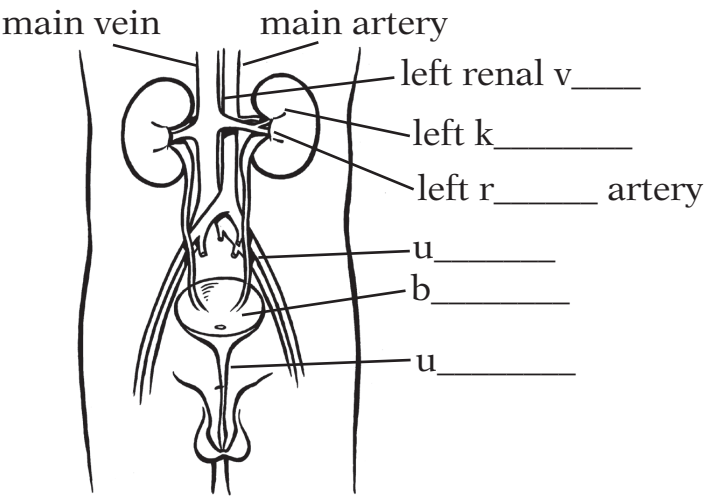
Human kidneys and excretion

The kidneys play a major role in removing waste from the human body. They filter blood through tight balls of capillaries, each called a **glomerulus**. The waste product that remains is called urea. Urea is **excreted** from the body in a liquid called urine.

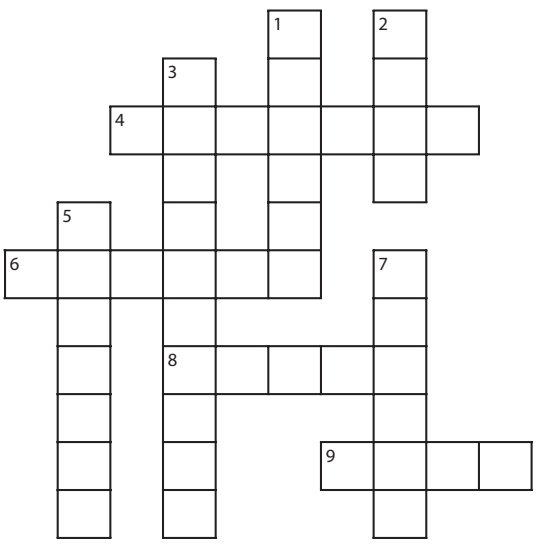
☞ **1** Use the word bank below to label the diagram.

Word bank

ureter bladder vein kidney urethra renal



2 Complete the crossword puzzle.



Across

4. Expansive organ that stores urine
6. Renal _____ takes blood to the kidney
8. Waste fluid
9. Renal _____ takes blood from kidney

Down

1. Organ that cleans blood
2. The end product of nitrogen excretion in mammals
3. Ball of specialised capillaries
5. Where urine is excreted from the body
7. Tube from kidney to bladder

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Plant responses to temperature

☞ **1** Read the information below, then underline any examples of how plants respond to temperature changes.

Oranges grown in cool regions have green skin while their fruit forms in the hot summer months. In autumn when the fruit ripens, the skin colour changes to orange. This colour change is due to the cooling air temperature. Ripe oranges grown in tropical climates normally stay green-skinned because there are no cooler months.

Plants are very sensitive to temperature changes and day length. **Deciduous** trees lose their leaves in autumn when the temperature drops and days are shorter. The trees become **dormant** and their **sap** doesn't flow. In countries where snowfall is frequent this leaf-fall and dormancy protects the trees from freezing. Bulbs such as tulips flower best when they have been through a period of chilling, as if packed under snow. In Australia these conditions are rare so commercial growers 'force' tulips into flowering by refrigerating them for several months.

2 Draw deciduous trees along the timeline, showing how they change throughout a typical year.

3 List two examples of deciduous trees.

_____

_____

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Teacher's notes – Hormones

Achieve!

Objectives

- To learn that our bodies constantly secrete hormones
- To understand that health problems occur if a hormone is not performing correctly
- To know that plants also produce hormones
- To understand that there are ethical and health problems associated with the abuse of hormones

Prior knowledge

Students should be aware of male and female hormones and their role in puberty.

Background

The endocrine system consists of glands and the hormones they produce throughout our body. These hormones are secreted into the bloodstream and coordinate growth and development, and the ways our bodies respond to various stimuli.

Starter activity

Ask students how they would respond if a wild lion, a poisonous snake or a masked gunman entered the room. What happens inside our body when we either fight or flee? List the physical responses. Talk about how the hormone adrenaline is responsible for these changes.

Activity sheets

The Activity sheet 'Insulin and diabetes' explains how the hormone insulin controls blood sugar. Students have a cloze exercise to complete detailing the role of insulin and how this relates to diabetes. There is also a graph to analyse, which shows the blood sugar levels of individuals with and without type 2 diabetes.

In the 'Action and adrenaline' Activity sheet, students unjumble sentences explaining how adrenaline affects our bodies when we undergo extreme emotions. There is also a flow chart to label and statements to match.

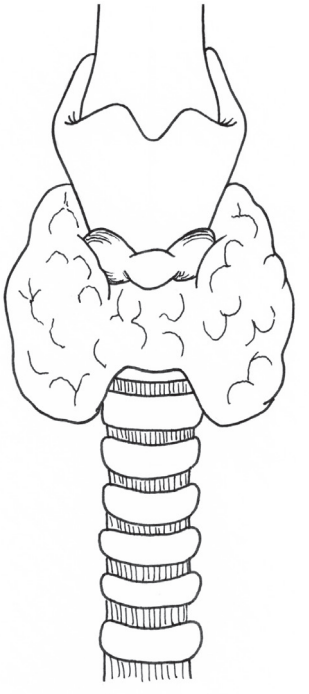
'Hormones in sport' explores the fact that some hormones are illegally used to enhance athletic performance. This Activity sheet has a cloze exercise to complete as well as a table to fill out regarding these banned substances.

The Activity sheet 'Thyroid gland' explains that the thyroid gland is a little-known gland that secretes a hormone controlling metabolism, growth and development. Students are asked to identify spelling errors in a passage about the thyroid gland. There is also a diagram to label. As an extra activity students can take notes from the passage.

The 'Plant hormones' Activity sheet looks at hormones in plants. This activity sheet outlines the five main types of plant hormone. Students complete a cloze exercise before matching pictures to the hormones they represent.

Extension ideas

Play 20 questions, getting the students to use a process of elimination to work out which hormone you are thinking of.



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Insulin and diabetes

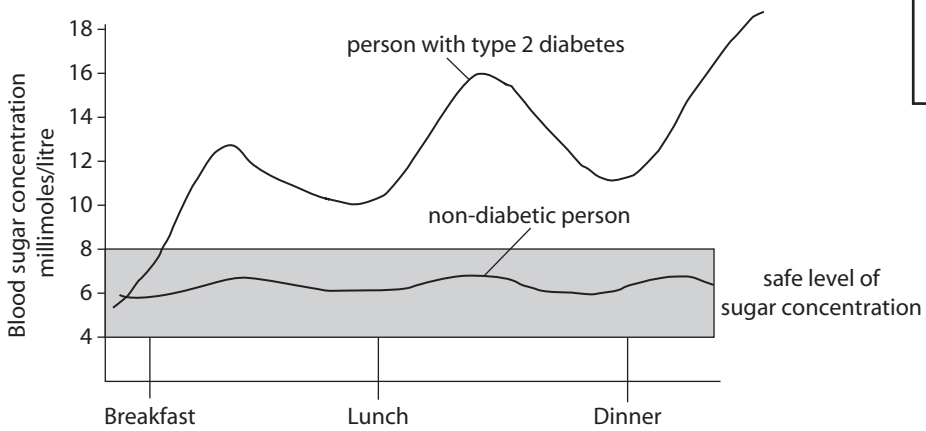
Word bank

hormone
coma
control
blindness
blood
enough
diabetes
glucose
insulin
diet
bodies
pancreas
vessels
damage
diabetes
exercise

☞ **1** Use the word bank to complete the passage below.

Glucose in the blood gives us energy. If a person doesn't have e_____ blood **glucose**, they can go into a c_____ and may die. Too much g_____ in the blood also causes problems. Over time it causes d_____ to the small blood v_____ in our body and can lead to b_____, nerve damage and **cardiovascular** disease.

The h_____ insulin is made by the p_____. Its job is to regulate the body's glucose levels. Some people's bodies don't make enough i_____. This problem is called type 1 diabetes. People with type 1 d_____ need regular doses of insulin each day to help c_____ their blood sugar. Other people find that as they age their b_____ sugar levels are hard to control. Their b_____ make insulin but it isn't doing its job effectively. This is called type 2 diabetes. In many cases this type of d_____ can be controlled with regular e_____ and a healthy d_____.



Blood sugar concentration is measured in millimoles per litre. The safe levels of blood sugar concentration are between 4 and 8, as shown on the graph.

2 Why are blood sugar levels so low for both people before breakfast? _____

_____

3 What happens to blood sugar levels soon after a meal? _____

4 Why does the person with type 2 diabetes have high levels of blood sugar throughout the whole day? _____

_____

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Action and adrenaline

The **hormone adrenaline** is released into the bloodstream when we face sudden emotional stress, such as fear or anger.

☞ **1** Unjumble the sentence parts and then rewrite the information about adrenaline in the correct order.

are found Adrenal **glands** of each kidney. on top

_____

adrenaline. the hormone They make

_____

a temporary effect Adrenaline has on the body.

_____

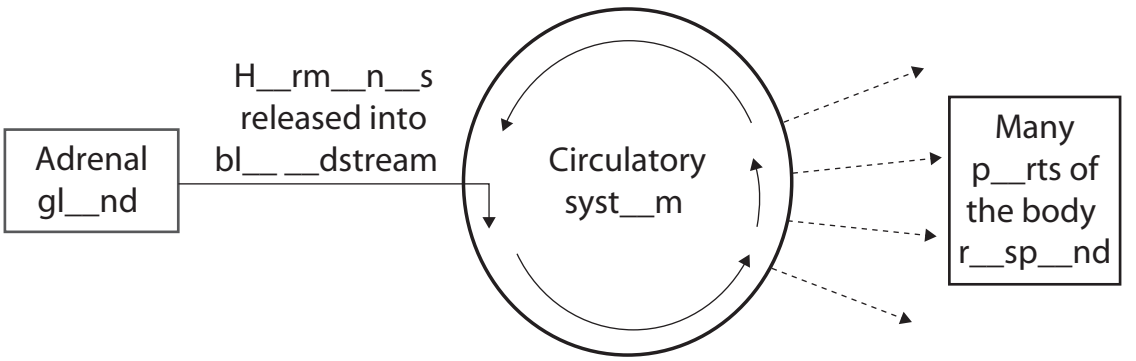
the muscles. increasing blood supply to for action by It prepares the body

_____

blood pressure, It also raises heartbeat and breathing rates.

_____

2 Complete the flow chart by adding the missing vowels (a,e,i,o,u).



3 Match the phrases below.

Adrenaline is produced	to react to sudden emotional stress.
When we are frightened	our blood pressure and heart rate.
Adrenaline prepares our bodies	adrenaline is released into our blood.
Adrenaline raises	in the adrenal gland.

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Hormones in sport

☞ **1** Use the word bank to complete the passage below:

Word bank
growth hormone body testosterone disorders banned deeper drug steroids male strength anabolic different hair athletes detect competitors side effects

Testosterone is a male h_____. It controls the development of m_____ **secondary sexual characteristics**, such as a d_____ voice and more body h_____ than in females. Anabolic s_____ are drugs derived from t_____. They are d_____ to the steroids that are used to treat asthma. For many years a_____ steroids were secretly used by some athletes and b_____ builders to increase muscle bulk, s_____ and **endurance**. These c_____ were considered to be d_____ cheats and the drugs were b_____. Today doctors warn against the use of anabolic steroids, as there are many s_____ effects. Another banned drug used by some a_____ is a growth hormone known as Human Growth Hormone, or hGH. This hormone regulates g_____ and is medically used to treat growth d_____ in patients. But some athletes think that hGH is just as effective as anabolic steroids, has fewer side e_____ and is harder to d_____ in drug tests.

2 Complete the table below.

	Hormone type	**Side effects include**
Anabolic steroids		Heart and liver damage, aggression, exaggerated secondary male sex characteristics
hGH		Heart and liver damage, abnormal bone growth in skull

3 What characteristics might a woman taking anabolic steroids develop?

_____

_____

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Thyroid gland

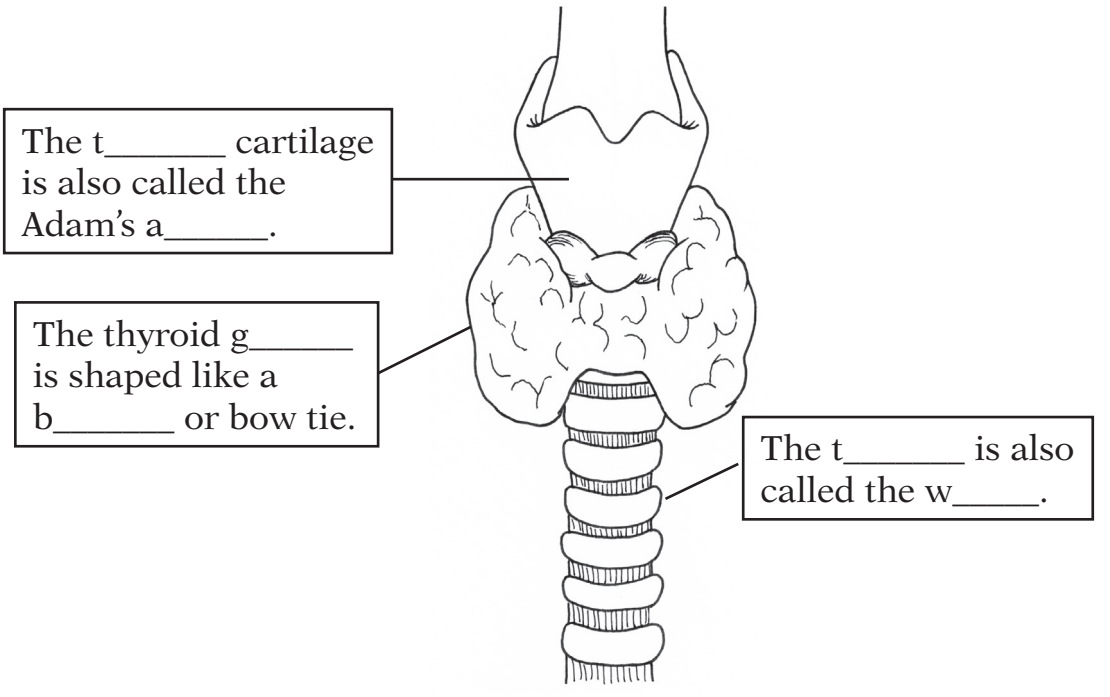
☞ **1** Ten words are spelt incorrectly in the passage below. Correct them and then re-read the text.

The tyroid gland is found in the neck in front of the winde pipe. It wraps around the trachae in two halves like buterfly wings. A normal thyroid gland cannot be seen, but when the gland is enlarged it forms a bulge just underneath the Adem's apple. The thyroid gland **secretes** thyroid hormones, which control the body's **metabolism** and heart rate. Babies and children need thyroid hermones for growth and brain development. The thyroid glande needs the element **iodine** to produce its hormones. Foods rich in iodine include iodised salt, seafood and plants grown in iodine-rich soil. People with a diet defishent in iodyne suffer an enlarged thyroid, witch is called a **goitre**.

2 Use the word bank to complete the diagram labels below.

Word bank

apple windpipe thyroid gland butterfly trachea



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Plant hormones

There are five main **classes** of plant hormones, some of which are made from a number of different chemicals. The classes are grouped according to similarities in their chemical structures and the effects they have on plant growth.

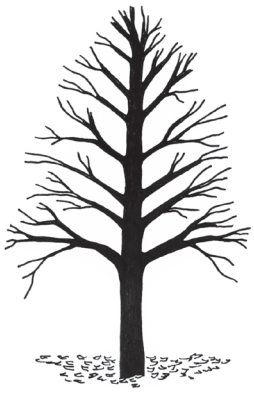
☞ **1** Use the word bank to complete the sentences below.

Word bank				
ABA	fruit	cell	leaf	shoot
division	growth	**stomata**	root	ripening

- **ABA** – plays a role in l_____ and seed **dormancy** by **inhibiting** growth; also in closing s_____ and slowing growth in mature leaves.
- **Auxins** – influence c_____ enlargement, bud formation and r_____ development.
- **Cytokinins** – influence cell division and s_____ formation. Work with auxins to affect stem and leaf growth, work with ethylene to promote f_____ and leaf drop.
- **Ethylene** – affects cell g_____ and cell shape in newly growing and dividing cells, affects fruit r_____.
- **Gibberellins** – reverses the effects of A___, promotes flowering and cell d_____.

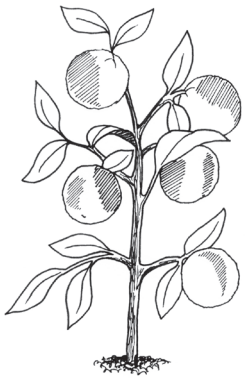
2 Draw a line linking each picture to the hormone it represents.

Auxins



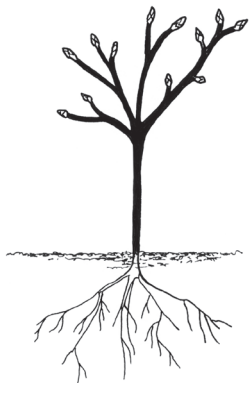
ABA

Cytokinins



Ethylene

Gibberellins



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Teacher's notes – Disease and deficiency

Achieve!

Objectives

- To know that plants and animals suffer from diseases and deficiencies
- To know that our bodies have built-in defences against disease
- To understand that vectors play a role in spreading some diseases
- To be aware that different pathogens are transmitted by various means
- To recognise that deficiency causes severe problems in both plants and animals

Prior knowledge

Students should be aware of basic hygiene practices and how these combat the spread of diseases like the common cold. They may also be aware of deficiencies in starving populations, or anaemia in people whose diets are low in iron.

Background

Diseases such as colds and flu are commonplace and our bodies have their own defences to fight these illnesses. There are many agents that cause disease, such as viruses, bacteria and fungi, and different diseases are transmitted in different ways. Plants also suffer from disease, but infection is often localised because transmission depends on close contact with other plants, or wind and insects to distribute pathogens. Nutrient deficiencies often show signs similar to disease. Both plants and animals are susceptible to deficiencies. Animals eat to provide their bodies with the necessary nutrients, whereas plants take their nutrients from the soil, as well as making some of their own through photosynthesis.

Starter activity

Ask students about strategies to avoid getting the flu. List these as a class. Ask them to describe what happens when we have the flu or a cold. What is the role of mucus? Why do we cough? Why do we develop a temperature? What are some other signs and symptoms of disease?

Activity sheets

The Activity sheet 'Human immune system' explains that our bodies have a built-in defence system. Students label a flowchart which details the front-line of the human immune system when combating disease organisms. There are also squashed sentences to separate out and examples to list in a table.

The 'Ross River virus' Activity sheet introduces the Ross River virus, an Australian disease spread by the mosquito. The mosquito is a vector, spreading disease from one organism to another. There are comprehension questions to complete and a flow chart to label.

The Activity sheet 'Devil facial tumour disease' explains how this disease currently affects the Tasmanian devil population. Using the notes about the disease, students should be able to expand the information into a paragraph. There are also true/false questions to test their understanding.

Plants also suffer from disease. The Activity sheet 'Plant diseases' presents three types of plant diseases. There are comprehension questions to complete and a table to fill out. As an extra activity, students could take notes from the passage.

The 'Vitamin D and sunlight' Activity sheet explains how our bodies synthesise vitamin D from sunlight. Notes are given on the topic and students have some higher order questions to consider. As an extra activity students could expand the notes to form a paragraph.

The Activity sheet 'Nutrient deficiencies in plants' presents an incomplete table listing deficiency symptoms in plants. Students need to fill in the missing vowels. Students also have squashed sentences to separate out and a plant deficiency to diagnose.

Extension ideas

Ask students to research a disease or deficiency. They could make a poster detailing how to prevent the illness/deficiency.

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Human immune system

☞ **1** Use the word bank to complete the flow chart.

First line of defence

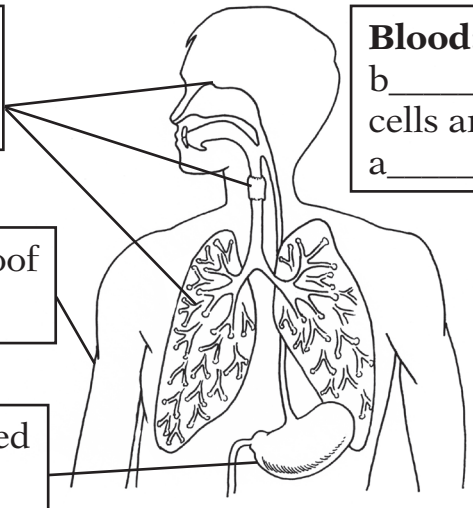
Respiratory s_____: dust and germs t_____ in sticky m_____.

Skin: w_____, germ-proof **self repairing** b_____.

S_____: many g_____ killed by e_____ and stomach a_____.

Second line of defence

Blood: white b_____ cells and a_____.



Word bank

Stomach
blood
barrier
mucus
germs
system
trapped
enzymes
antibodies
waterproof
acid

2 Separate the words below to make sentences.

Anillnessissometimescalledaninfection.

_____

Lymphnodesarealsocalledlymphglands.

_____

Chemicalsproducedduringaninfection arecalledantibodies.

_____

Enzymesandacidarereleasedinthestomach.

_____

Lymphnodesproducewhitebloodcells.

_____

3 The following bodily reactions occur as part of our body's defences. List them in the table as either first- or second-line defences.

fever scab forming sore glands blood clotting coughing up mucus

First line of defence	Second line of defence

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Ross River virus

Many diseases are spread by contact between people but some are spread by **vectors**. Vectors are disease-carrying organisms. The mosquito is a common vector that can spread serious diseases such as malaria and dengue fever. In Australia the Ross River virus is spread by mosquitoes. The virus mostly infects people in rural areas, and causes a **flu-like** illness with a rash, joint pains and fever. Sometimes joints can become swollen. The illness can last for several weeks, making the patient feel weak and tired. Fortunately it is not fatal, but there is no specific treatment.

☞ **1** List the diseases that can be spread by mosquitoes. _____

_____

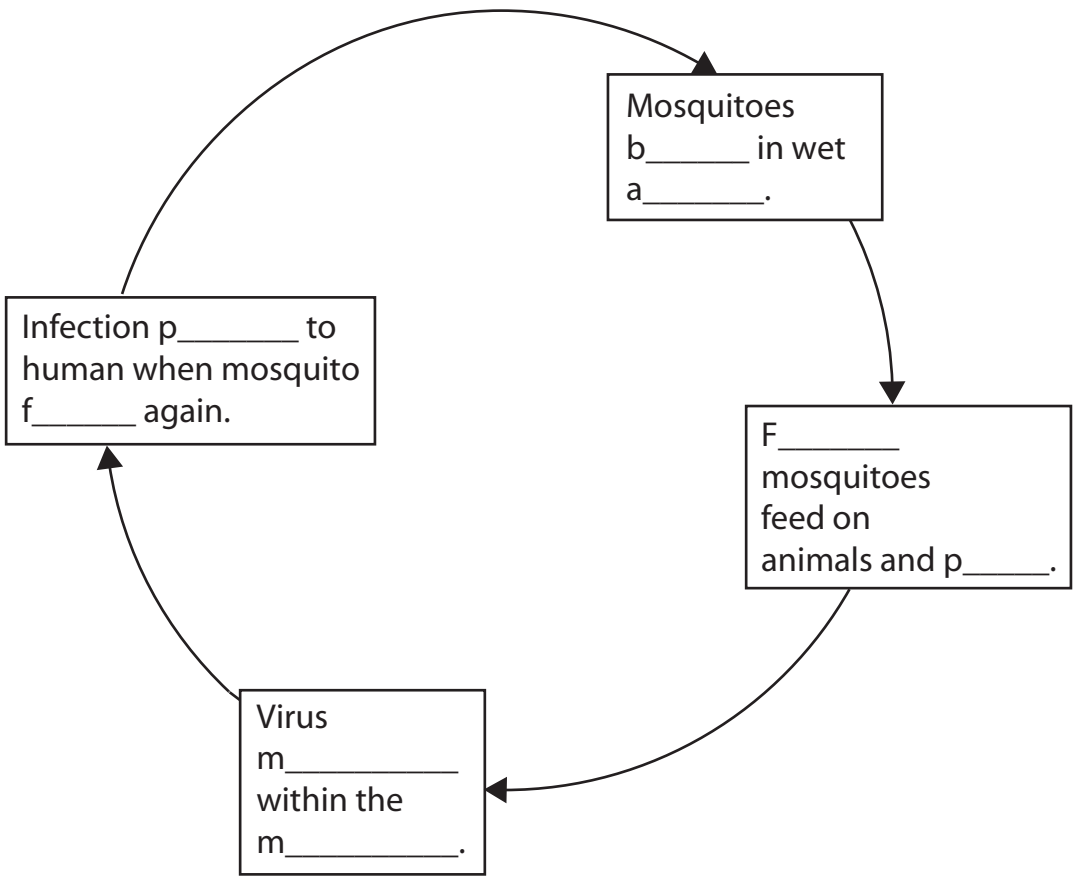
2 How can people protect themselves from being bitten by mosquitoes? _____

_____

3 Which of these is another word for fatal:

a) debilitating **b)** die **c)** death **d)** deadly

4 Complete the flow chart of the infection cycle.



Word bank

passes
Female
breed
areas
feeds
mosquito
multiplies
people

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Devil facial tumour disease

☞ **1** Use the notes below to write a paragraph on devil facial tumour disease.

- Tasmanian devils are **endangered**
- many suffer from devil facial tumour disease (DFTD)
- DFTD is a rare **contagious** cancer
- starts as small lumps and cuts around the mouth
- develop into large **tumours** around the face and neck
- tumours spread throughout the body
- infected devils can't eat properly; they weaken and die
- spread by infected devils biting other devils
- sightings of devils have declined by over 80% since DFTD was first noted in the 1990s

_____

_____

_____

_____

_____

2 Why would a Tasmanian devil bite another devil? _____

_____

3 Write whether the following statements are true or false.

a) Tasmanian devil numbers have declined rapidly. _____

b) DFTD is spread by coughing. _____

c) DFTD is a virus. _____

d) Many devils starve to death because they can't eat. _____

e) All cancers are contagious. _____

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Plant diseases

Like animals, plants also suffer from disease. **Fungal** diseases such as mildew and brown rot are common in vegetable and fruit crops. The fungus spreads from plant to plant through tiny fungal spores which float in the air. The spores thrive in warm, moist conditions. **Bacterial** diseases such as bacterial blight are also airborne and thrive in **humid** crowded conditions, such as among potted glasshouse plants. Plant diseases are also spread by insects. Bunchy top is a serious **viral** disease affecting bananas. It is spread by banana **aphids**, which carry the disease from plant to plant.

☞ **1** The three types of plant pathogens are:

_____ _____ _____

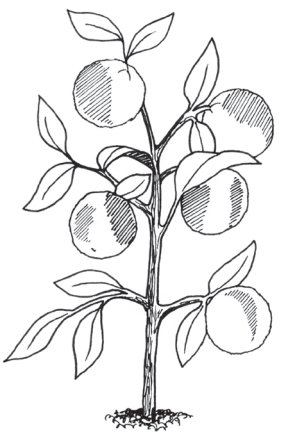
2 Humid means:

a) cold and wet **b)** warm and wet **c)** warm and dry **d)** cold and dry

3 Should potted plants be packed together tightly in a greenhouse?
Why or why not? _____

4 Use the information in the text above to complete the following table. Draw lines to join each plant illustration to the disease that affects it.

	Brown rot	Bunchy top	Bacterial blight
Pathogen (organism that produces the disease)			
Example of affected species			
How disease is spread			



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Vitamin D and sunlight

Many Australians avoid the sun due to the threat of skin cancer. However, some exposure to sunlight is actually good for us. Read the notes below and then answer the questions.

Melanin

- Found in skin cells
- Colours our skin
- Acts as a natural barrier to **UV rays**
- Dark-skinned people have lots of melanin
- Fair-skinned people do not have much melanin

Vitamin D deficiency (also known as rickets)

- Caused by a lack of sunlight
- Leads to bones softening; children's legs can grow bent
- Common in industrial Europe centuries ago
- Occurs in Australia today in elderly populations and dark-skinned people in cold regions
- Can be cured by moderate **exposure** to sunlight

☞ **1** Why do dark-skinned people who live in cold climates often develop a vitamin D deficiency? _____

2 When babies are born their skin colour is paler than their parents. This allows babies and children to make lots of vitamin D. Why do you think this is? _____

_____

3 Are these statements true or false?

a) We can only get Vitamin D from our diets. _____

b) People with pale skin don't need any vitamin D. _____

c) Melanin protects us from UV rays. _____

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Nutrient deficiencies in plants

Plants take up **inorganic nutrients** through their roots. Some soils don't have a good range of nutrients. When this happens the soil is nutrient deficient, and the plants cannot get all the nutrients they need. The table below lists symptoms common in plants that are **deficient** in major nutrients.

☞ **1** Fill in the missing vowels (a, e, i, o, u).

	Nitrogen	**Phosphorus**	**Potassium**
Needed for	Regulates ch__m__c__l reactions, also stem and l__ __f growth	Root formation, fl__w__r__ng, fruiting and ripening	Fr__ __t__ng, flowers and gr__w__ng tips, also for ph__t__synth__s__s at low light levels
Deficiency signs	Leaves turn pale gr__ __n to yellow	St__nt__d growth, dark leaf c__l__ __r, poor flowering __nd fruiting	Spotted older l__ __v__s with dry edges. Weak stems and uneven fr__ __t ripening.
Remedy	Urine, all types of animal m__n__r__, fish and f__ __th__rs	Bonemeal, b__rd manure	S__ __w__ __d, wood ash

2 Separate the words below to make sentences.

Plantswithyellowleavesmayhaveanitrogendeficiency.

_____

Addingbirdmanuretothesoilwillincresenitrogenlevels.

_____

Heavyraincanwashnitrogenoutofsandysoils.

_____

3 What deficiency would the following plant have?

_____

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Teacher's notes – Medicine and technology

Achieve!

Objectives

- To learn that vaccinations prevent the spread of serious diseases
- To realise that there are a variety of types of medical imaging
- To know that technological advances can monitor health on a regular basis
- To be aware of the basic instruments used in measuring cardiovascular health

Prior knowledge

Students should be aware of a range of medical technology, from stethoscopes and thermometers to X-rays.

Background

Medical technology has come a long way in the last hundred years. Electronic devices, implants and a variety of imaging techniques are all commonplace today, but traditional instruments such as the stethoscope are also still used.

Starter activity

Get students to brainstorm all the technologies they can think of relating to medicine. How are our temperatures taken? What about imaging of foetuses and broken bones? How is our heart function measured? Students will have seen many types of equipment but may not know the technical names.

Activity sheets

The Activity sheet 'Vaccinations' explains that vaccinations were first developed over two hundred years ago. Students read a passage introducing the topic of vaccinations and how they are taken. This is followed by some comprehension questions and some phrases to match.

The 'Monitoring health' Activity sheet introduces some chronic health conditions, such as type 1 diabetes and arrhythmia, which need constant monitoring. Implants are now available that can monitor these conditions and maintain correct insulin levels and pulse rate. There are two passages to complete, followed by some higher order questions to consolidate understanding.

In 'Detecting kidney stones', students learn how kidney stones are commonly detected using imaging from X-rays, ultrasounds or CAT scans. Students read notes about these three techniques before answering some comprehension questions and labelling a diagram of the excretory system. Some true/false statements conclude the activity.

Stethoscopes and sphygmomanometers are common medical instruments used to measure blood pressure. In 'Measuring blood pressure', students are given a cloze exercise to complete, true/false questions and a diagram to label in order to become more familiar with these hard-to-pronounce technologies.

Extension ideas

Ask students to produce a poster about a piece of medical technology, including a picture or diagram and an explanation of its use.

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Vaccinations

Vaccines are liquids that contain either dead **germ** particles or harmless live germs. The germs used to make a vaccine are similar to the germs that cause the **disease** being vaccinated against. When a vaccine enters the bloodstream, the body reacts as if it were actually suffering from an infection and makes **antibodies**. The antibodies stay in the blood for a long time, making the body **immune** to a real infection.

Scientists have produced vaccines for a range of deadly diseases such as tetanus, diphtheria and measles. Most vaccinations are injected but some, such as the polio vaccine, are swallowed.

☞ **1** If vaccines contain germ particles why doesn't the vaccinated person become sick with the disease? _____

_____

_____

2 Another word for immune is:

a) susceptible **b)** reluctant **c)** vulnerable **d)** resistant

3 How are vaccines given to humans? _____

4 List as many other kinds of vaccinations as you can think of.

_____

_____

5 Match the phrases below.

Before vaccinations, smallpox	are taken orally.
Smallpox was a	occurred in Somalia in 1977.
Vaccines have been	killed thousands of people every year.
Some vaccinations	against serious diseases.
Even pets are vaccinated	highly infectious disease.
The last case of smallpox	developed for many diseases.

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Monitoring health

Some technology has been developed to monitor a patient's health on a minute by minute basis. Two examples are presented below.

☞ **1** Add the missing vowels (a, e, i, o, u) to complete the passage.

Some people s__ff__r from **arrhythmia**, which is when the h__ __rt may beat too slow, too fast or with an irregular rhythm. It can c__ __s__ fainting, shortness of breath and fatigue. **Surgeons** often recommend a p__c__m__k__r implant for arrythmia sufferers. The pacemaker is a sm__ll device that is placed in the patient's chest or abdomen to c__ntr__l abnormal heart rhythms. It uses electrical s__gn__ls to prompt the heart to b__ __t at a normal rate.

2 Use the word bank to fill out the passage below.

Word bank

eats	**insulin**	easy	bloodstream	**diabetes**
under	pumps	device	program	monitors

People with type 1 **d**_____ now have the option of using an **i**_____ pump. The d_____ is very small and is partly placed u_____ the skin of the abdomen. It m_____ glucose levels in the blood and p_____ a base rate of insulin into the b_____. When the diabetic person e_____ a meal, they are able to p_____ the device to pump extra insulin into their bloodstream. The pump is discreet and e_____ to use.

3 Why might young diabetics find the insulin pump more popular than regular daily injections of insulin?

_____

4 Both the pacemaker and the insulin pump are implanted under the skin of the patient. How is this an advantage? _____

_____

5 What is the advantage of constant monitoring with arrhythmia and type 1 diabetes?

_____

_____

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Detecting kidney stones

Sometimes substances that are normally dissolved in urine separate out as a solid, making hard little stones called **kidney stones**. The stones move through the excretory system, scratching and blocking the thin tubes of the kidney, ureter and urethra. It is very painful. There are three ways for doctors to see if a patient has kidney stones.

Standard X-ray – low doses of **radiation**

- Uses a photographic negative
- Detects bones and calcium stones
- Cheap and quick

Ultrasound – sound waves passed over the skin near kidneys, ureters and bladder

- Image presented on screen
- Detects both calcium and non-calcium stones
- No radiation

CAT scans – an X-ray tube that creates many images

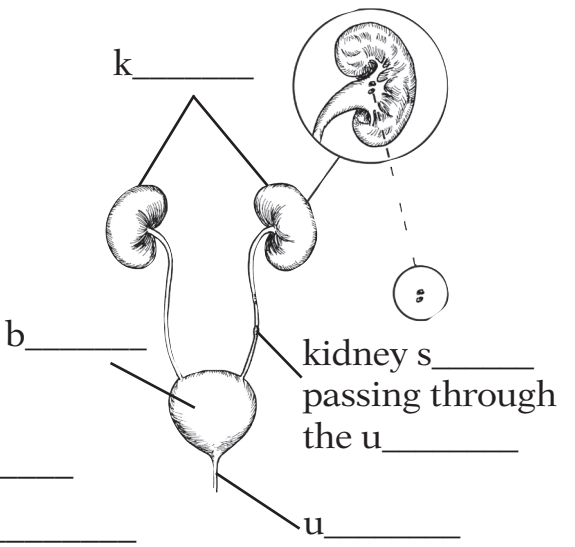
- Sometimes detects non-stone causes of severe pain
- Sensitive but expensive
- Uses more radiation than a conventional X-ray.

☞ **1** Which method is best for someone who shouldn't be exposed to radiation, such as a pregnant woman? _____

2 Which method is best for someone in severe pain? _____

3 Use the word bank to label the diagram.

Word bank				
kidneys	ureter	stones	urethra	bladder



4 Are these statements true or false?

a) Kidney stones are only found in the kidney. _____

b) A standard X-ray is cheap and quick. _____

c) CAT scans can only detect kidney stones. _____

d) An ultrasound does not use radiation. _____

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Measuring blood pressure

Word bank

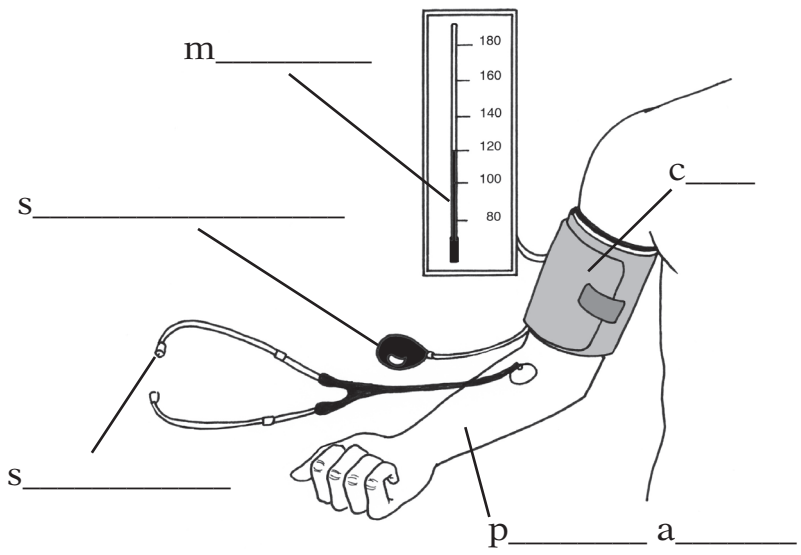
pressure
diastolic
body
maximum
heart
artery
two
sphygmomanometer
level
listens
mercury
stethoscope
pumped

☞ **1** Use the word bank to complete the passage below.

Measuring blood pressure tells doctors a lot about a patient's h_____ and the condition of their blood vessels. Traditionally t____ important pieces of technology were always used for measuring blood pressure – the **stethoscope** and the **sphygmomanometer**.

The s_____ has a cuff, two air tubes and a reservoir of m_____ in a vertical glass column. When air is p_____ into the cuff, the pressure is shown by the mercury l_____ in the column. The doctor puts the s_____ to the patient's arm and l_____ to their pulse. The **systolic** pressure is when the doctor first hears the pulse. It is the m_____ pressure in an a_____ when the heart beats, pumping blood through the b_____. The d_____ pressure is when the doctor can't hear the pulse any more. This is the lowest p_____ in an artery. Nowadays blood pressure is monitored electronically using a cuff and a special machine, but the traditional method works just as well.

2 Use the word bank to label the picture below.



Word bank

mercury
cuff
stethoscope
sphygmomanometer
patient's arm

3 Decide whether these statements are true or false.

The stethoscope has a cuff attached to it. _____

The cuff pressure is shown on the mercury column. _____

Diastolic pressure is measured when no pulse can be heard. _____

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Teacher's notes – Populations

Achieve!

Objectives

- To devise ways of counting populations within a defined space
- To recognise that many factors impact upon populations
- To understand that population growth is not sustainable when habitat is destroyed
- To recognise that populations fluctuate and models can be used to determine population growth
- To understand that some species populations and their roles are not yet properly understood

Prior knowledge

Students should be aware that populations change due to births, deaths and migration factors.

Background

Biologists are interested in counting populations to determine the health of a species in any given space. The method of counting differs depending on how easy individuals are to count. Once numbers are recorded the sustainability of a species can be seen over time. Factors such as habitat destruction, pollution, disease, changing weather conditions and competition can all impact on populations. These impacts can be expressed as quantitative data if populations are monitored.

Starter activity

Select an animal that is common in the area. Examples could be possums, mosquitoes, crickets, flies, jellyfish, cane toads or mice. Ask students to determine how the species could be counted. What area would be sensible to sample and why? How can sample numbers be extrapolated to give a plausible population count? Encourage students to make simple mathematical predictions. Explain that this is the basis for mathematical modelling of populations.

Activity sheets

The Activity sheet 'Murray cod' looks at the many factors that affect population numbers of the Murray cod. Students can expand the notes to write a passage on the topic. There are also higher order questions to answer and a table to fill out, listing the factors that cause population changes.

The 'Monitoring populations' Activity sheet explains that biologists need to understand the health of a species in a place. Students fill in a table outlining ways to count various organisms in a place. Students need to choose the best option and explain why they made the choice. They are also asked to devise a way of counting a particular weed population of the school.

'Koala numbers' looks at why koala populations are decreasing on the mainland, but have swollen to unsustainable numbers on Kangaroo Island. Students complete a cloze exercise on the topic then draw a graph. They need to determine whether their graph is a plausible model for predicting population growth.

The 'Kangaroo stomach worms' Activity sheet presents students with a current biological debate. Are the worms inside a kangaroo's stomach beneficial symbiotic organisms or antagonistic parasites? Ensure students understand that much of the behaviour and physiology of Australian animals is still being researched. Students write their own passage based on the notes provided, then there are statements to match and a table to fill out.

The Activity sheet 'Monocultures' explains that much of modern agriculture uses monocultures. Students read a passage about monoculture crops such as wheat, filling in the missing vowels before answering some true/false questions and analysing a graph.

The 'Crown-of-thorns starfish' Activity sheet presents a passage and comprehension activities about the Crown-of-thorns starfish. There are also statements to match. Students can list the main points in note form as an extra activity.

Extension ideas

Ask students to devise a board game based on populations within an ecosystem, where numbers are gained or lost.

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Murray cod

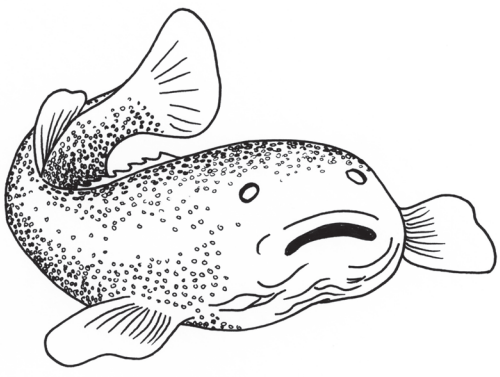
☞ **1** Read the notes below about the **declining population** of the Murray cod, then write a paragraph summarising the reasons for its decline.

- live in the waters of the Murray-Darling Basin
- are **territorial**, aggressive and live under submerged logs
- mature at approximately 5 years or 50 cm long
- can grow to over 1 metre and live for over 70 years
- **spawn** in warm waters
- are highly prized by fishermen
- population is in decline due to overfishing, clearing of submerged logs and drought periods that shrink waterholes, leading to greater competition for food and habitat
- dams, levees and irrigation limit species movement, which in turn limits **genetic diversity** at spawning due to in-breeding
- cold water released from dams prevents spawning

2 To help maintain population numbers, scientists have bred Murray cod in **hatcheries** and released them into the wild. There is also a minimum catch size of 50 cm long. Both of these strategies have had limited success. Why do you think this is?

3 List the factors that increase and decrease numbers in the Murray cod population.

Numbers increase	Numbers decrease



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Monitoring populations

To compare populations biologists need to look at **densities**. Density refers to the number of individuals of a species per unit area. For instance, species such as kangaroos and weeds can be counted on a per hectare basis. However, it's not possible to count every individual in a population. Individual animals can move or be hidden from view and some weeds grow in clumps or from underground shoots. To combat this, animals can be counted per nesting site or burrow. Clumping plants can be counted as the total weight of the population per unit area. Insects and **microorganisms** might be best expressed as number of individuals per unit of volume. For example water bugs could be counted per litre of pond water.

☞ **1** Read the passage above and underline the different methods that can be used to count populations.

2 Fill in the table below, circling the method you would use to count each population and giving a reason for your answer.

Species	Method of counting	Why
rabbits in a paddock	• individuals per hectare • burrows per hectare	
mosquito larvae in a tank	• individuals per tank • individuals per litre of water	
fish in a large dam	• individuals per square metre of water • kilograms per hectare of water	
thrips in a rose bush	• individuals per flower • grams per rose bush	
capeweed in a garden	• individuals per square metre • kilograms per garden bed	

3 How would you count the number of **broad-leafed** weeds on the school oval or garden beds? _____

_____

4 What is the best measurement of area to use? How do you reach the final number?

_____

_____

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Koala numbers

Word bank

stripped
koalas
falling
adult
tourists
Australia
predators
fur
ecosystem
conservationists
population
cars
numbers
habitats
conserve

☞ **1** Use the word bank to complete the passage below.

Koalas are cute and t_____ love them. But almost a century ago they were hunted for their f_____. In the 1920s the South Australian government tried to c_____ koala numbers by introducing 18 adult k_____ to Kangaroo Island. Because there were no p_____, and they were not being hunted, the koala population increased steadily. In 2001 koala n_____ on the island were estimated at 27 000. There were so many koalas that they s_____ and eventually killed the manna gum trees they lived on, damaging the **e**_____ of the island. In response, the government started **de-sexing** some a_____ koalas and removing others to the mainland.

Elsewhere in A_____, koala numbers are f_____ as the koalas are threatened by logging, foxes, dogs, c_____ and **urbanisation**. Some c_____ are against the mainland resettlement program. They believe that adding more koalas puts pressure on the shrinking mainland h_____. They also believe that random de-sexing can limit the **gene pool** in the already in-bred p_____ of Kangaroo Island.

2 Use the information above to plot a graph showing the increase in the koala population on Kangaroo Island from the 1920s to 2001.

3 According to the graph, how many koalas do you think there would be in 2030 if numbers weren't controlled? _____

4 Do you think this modelling would be accurate? Why/why not? _____

_____

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Kangaroo stomach worms

☞ **1** Use the notes below to write a paragraph about the digestive system of a kangaroo.

- kangaroos eat grass but can't digest it
- their stomachs are full of long thin worms and bacteria
- the stomach worms digest the grass
- kangaroos digest the stomach worms' waste
- some people think the worms are parasites
- some people think kangaroos need the stomach worms to digest the grass
- both sick and healthy kangaroos have the worms

A **parasitic** relationship is when one organism lives off another, causing it harm. A **symbiotic** relationship is when two different organisms both benefit from their relationship and neither suffer.

☞ **2** Match the phrases below.

A parasite takes nutrients	lots of stomach worms.
The stomach worms can't	from the host.
A healthy kangaroo has	exist outside of the kangaroo.

3 Are the worms parasitic or symbiotic? List the possible evidence to support each argument.

Parasitic	Symbiotic

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Monocultures

☞ **1** Add the missing vowels (a, e, i, o, u) to complete the passage below.

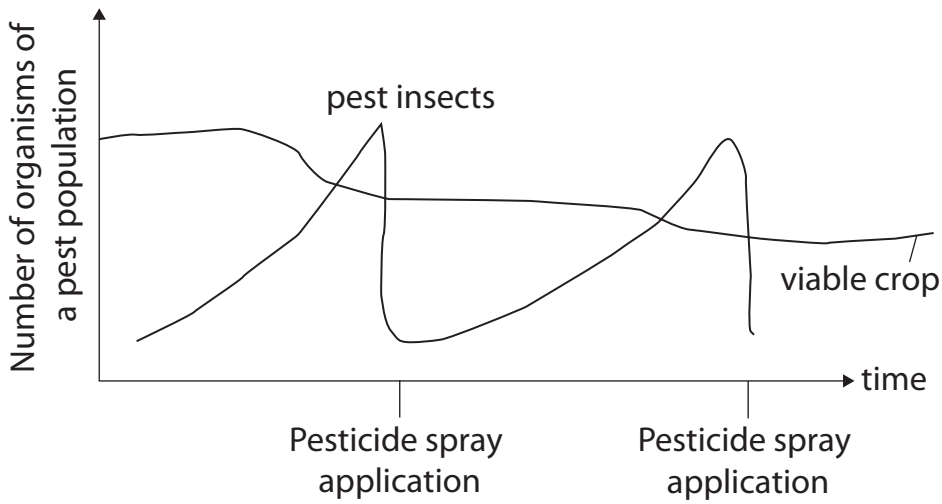
Throughout the world, much of our f__ __d is grown in **monocultures**. This means that one cr__p is grown over vast areas, and any c__mp__t__ ng weeds or other plants are removed by the farmer. Think of the huge paddocks of wh__ __t in Australia as an example, or rice in __s__ __. M__n__c__lt__r__s are easy to manage. Uniform q__ __nt__t__ __s of fertiliser are applied and **irrigation** is used if needed. The crop can be harvested without any c__nt__m__n__t__ __n from other species. However, monocultures are highly susceptible to __tt__ck from pests and diseases. This is because they are an enormous food source that l__cks the d__v__rs__ty and r__s__st__nc__ that would be found in a n__t__r__lly occurring environment.

2 Are these statements true or false?

Monocultures are resistant to pests. _____

Wheat is grown as a monoculture in Australia. _____

Monocultures need protection from pests and diseases. _____



3 Examine the graph above. Why do insect numbers rise rapidly? _____

_____

4 Why does the farmer need to spray again? _____

_____

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Crown-of-thorns starfish

Crown-of-thorns starfish are large starfish that eat **coral polyps**. In some years large outbreaks of the starfish occur on Australia's greatest tourist attraction, the Great Barrier Reef. The females produce approximately 60 million eggs a year. The eggs drift for months in the water as **zooplankton**, feeding on microscopic **algae**. In an outbreak year more of the eggs reach maturity because more nutrients have washed into the sea, creating more algae for the zooplankton to eat.

Once they grow larger they settle on coral. Adult starfish can reach 30 cm in diameter. They are covered in sharp poisonous spines that can pierce clothing. A single starfish can move up to 20 m per hour, searching for food, and in a year they can eat around 6 m^2 of living coral reef. As adults they have few predators.

☞ **1** What factors make the crown-of-thorns starfish such an effective predator?

_____

_____

2 Why do you think biologists want to control starfish outbreaks on the Great Barrier Reef?

_____

_____

3 As more humans farm coastal areas, more **nutrients** make their way into the sea, boosting algae populations. How does this contribute to increased crown-of-thorns starfish numbers?

_____

_____

4 Match the phrases below.

Destroyed coral populations	by injections of Sodium bisulphate.
Adult starfish are controlled	few natural predators.
Crown-of-thorns starfish	take ten years to recover.
This starfish has very	have up to 21 arms.

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Teacher's notes – Basic genetics

Objectives

- To understand the differences between mitosis and meiosis
- To know that DNA is the basis of genetic science
- To be aware of dominant and recessive characteristics
- To know the difference between genotype and phenotype
- To be able to predict the outcome of a simple genetic cross

Prior knowledge

Students should be aware that their individual characteristics are inherited from their parents. These include things that can be seen, such as hair and eye colour, and things that cannot be seen, such as blood type.

Background

Scientists have understood the basics of genetics since Mendel's work with pea plants, showing that some characteristics are dominant over others. However, this branch of science really took off after Watson and Crick discovered the structure of DNA molecules in the 1950s. Now DNA is talked about in fields as diverse as law, medical science and anthropology.

Starter activity

Explain the difference between attached and free lobed ears. Ask students to examine one another's ears and record the proportions of each type within the class. Explain that free lobes are dominant to the recessive attached ears.

Activity sheets

The Activity sheet 'Meiosis vs mitosis' explains how these two processes relate to cell division and replication. Students are given a passage outlining the differences between mitosis and meiosis. They need to choose the correctly spelled word in each pair before completing a table to demonstrate their understanding.

The 'DNA discoveries' Activity sheet introduces scientists Watson and Crick, who discovered the structure of the DNA molecule. Students are given a passage outlining the discovery and they need to underline anything to do with DNA structure. There is also a small cloze exercise and some true/false questions to consolidate their understanding.

The Resource and Activity sheets 'Mendel's pea plants (1) and (2)' introduce students to Mendel, who was the first person to fully study generations of plants and formulate the laws of inheritance. The Resource sheet outlines Mendel's procedure. Students should refer to this page when completing the Activity sheet. The Activity sheet has a small cloze exercise and a jumbled version of Mendel's experiment to sort. There are also some questions about pea plant phenotypes.

'Genes and chromosomes' presents a crossword puzzle introducing key terms relating to genetics. There are also some true/false questions to answer.

The Activity sheet 'Roll your tongue' explains that the ability to roll one's tongue is governed by genetics. Students fill out punnet squares and answer comprehension questions to predict this ability in various generations.

Extension ideas

Ask students to research a genetic topic to present to the class, such as coat colour in guinea pigs or cats, or genetic conditions such as haemophilia or albinism in people.

Selected answers

Genes and chromosomes Across: **1** chromosomes, **3** parent, **4** small, **7** dominant, **9** genes, **10** hybrid, **11** sex. Down: **2** heredity, **3** peas, **5** Mendel, **6** law, **8** helix

Roll your tongue **1** RR & Rr; **2** (i) RR x RR (ii) Rr x Rr (iii) RR x Rr; **3** (i) all rolling, (ii) ¾ rolling and ¼ non rolling (iii) all rolling; **4** rr x rr; **5** all rr; **6** non rolling.

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Meiosis vs mitosis

☞ **1** Circle the correct word in each pair.

Mitosis is when a nuecleus/nucleus divides and pairs of **chromosomes** are reproduced. Each **daughter cell** has the same number of chromosomes/cromosomes as the **parent cell**. This is called the **diploid** number/numbre of chromosomes. Mitosis takes place in sells/cells throughout the body or plant wherever growth occurs. Once an organist/organism has reached maturity mitosis/mytosis is rare, except where cells are being replaced.

Meiosis happens in the reproductive organs/oargans of sexually reproducing/reproduce organisms. Meiosis is when a nucleus divides/divide and pairs of chromosomes are separated, forming two new cells. Each new cell is called a **gamete**. Grametes/gametes each have halve/half the number of chromosomes as the parente/parent cell. This is called/cold the **haploid** number of chromosomes.

Fertilisation occurs/ocurs when the gamete from the mail/male fuses with the gamete from the female. The combined/combine gametes each contribute their harploid/haploid number of chromosomes, creating/creates an individual with a diploid number of chromosomes.

2 Fill out the tables below.

	Mitosis
Where it occurs in plants	
Where it occurs in animals	
Number of chromosomes compared to parent cell	

	Meiosis
Where it occurs in plants	
Where it occurs in animals	
Number of chromosomes compared to parent cell	

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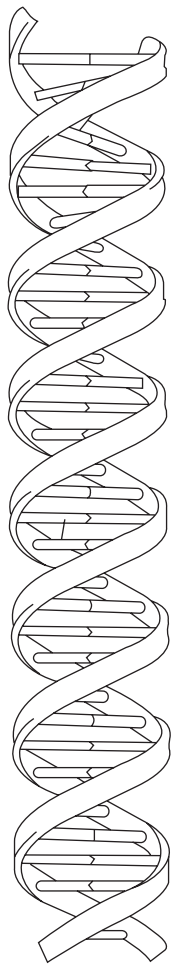
DNA discoveries

DNA stands for deoxyribonucleic acid. It is found in the **nuclei** of cells and contains genetic material called **genes**. Since the 1940s scientists have known that DNA is the **molecule** of life, but they didn't know what the molecule looked like. They knew DNA consisted of four **bases**: adenine (A), thymine (T), guanine (G) and cytosine (C). These bases joined together in pairs; A always pairs with T, and G always pairs with C. Scientists also knew that the structure had a phosphate backbone but they weren't sure how the bases formed along it. In 1953 two scientists named Watson and Crick proposed that the bases are paired like rungs in a twisted ladder that attaches to the phosphate backbone. The formation is called a **double helix**.

☞ **1** Read the passage above and underline any information about the structure of a DNA molecule.

2 Use the word bank to complete the sentence describing the illustration.

Word bank				
bases	outside	phosphate	molecule	four



The p_____ backbone is on the o_____ and the f_____ different b_____ are on the inside of the DNA m_____.

3 Decide whether these statements are true or false.

a) Watson and Crick were the only scientists working on DNA. _____

b) Watson and Crick discovered the DNA double helix. _____

c) DNA has four bases: A, T, G and C. _____

d) The phosphate backbone is on the inside of the DNA molecule. _____

e) G and T always pair up together. _____

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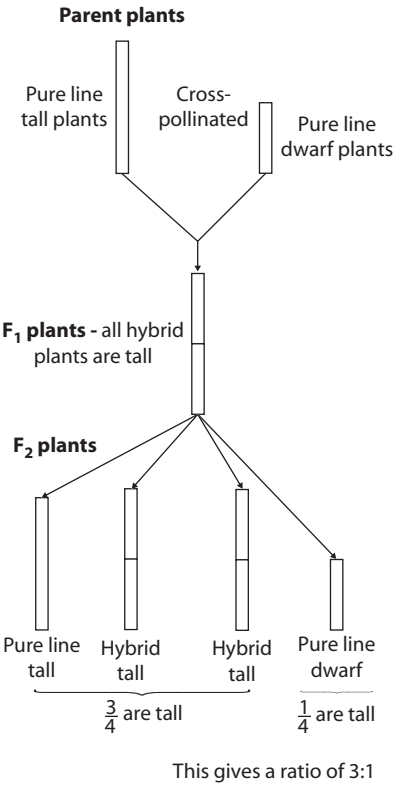
Mendel's pea plants (1)



This gives a ratio of 3:1

Gregor Mendel bred pea plants to study the laws of **heredity**. Peas grow quickly and there are many types. Mendel chose a pure line of tall peas to cross pollinate with a pure line of dwarf peas. He called the seeds from this pairing the F_1 (first generation) plants, or the F_1 generation. They were called **hybrids** because they grew from a cross between two genetically different plants. All the F_1 plants were tall.

- Mendel said that tallness **dominated** 'dwarfness'.
- Mendel called the tall gene **T** and the dwarf gene **t**.

Next, Mendel pollinated all the F_1 plants with one another. He collected their seeds and planted them. The plants that grew were called the F_2 plants or F_2 generation. Some F_2 plants were dwarf but many were tall. From this, Mendel concluded that dwarfness was caused by a **recessive** gene.

Punnet squares are drawn up to predict the **phenotype** of individuals from a genetic cross.

- The **genotype** refers to the individual's genes.
- The phenotype refers to the individual's appearance.

Below is a punnet square which shows the **genotypes** of the peas in the F_1 cross.

	T	**t**
T	TT pure line tall plants	Tt hybrid tall plants
t	tT hybrid tall plants	tt pure line dwarf plants

¼ TT
¼ Tt = ¾ tall plants
¼ tT

¼ tt = ¼ dwarf plants

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Mendel's pea plants (2)

☞ **1** Use the word bank to complete the passage below.

Word bank
Mendel dwarf tall plant represented dominated seeds

When M_____ cross-pollinated tall and d_____ plants, he planted their s_____ to produce the F_1 generation. Every p_____ from the F_1 generation was t_____. Mendel declared that tallness had d_____ the **recessive** dwarfness. He r_____ tallness as 'T' and dwarfness as 't'.

2 Order the steps Mendel took to study the laws of heredity.

	Step number
Plant F_1 seeds and watch them grow into F_2 plants	
Cross-pollinate tall and dwarf plants (parents)	
Plant parent seeds and watch them grow into F_1 plants	
Collect seeds of parent plants	
Cross-pollinate all F_1 plants then collect their seeds	

3 What is the phenotype of a pea plant with the genotype **Tt**? _____

4 What is the phenotype of a pea plant with the genotype **TT**? _____

5 What is the phenotype of a pea plant with the genotype **tt**? _____

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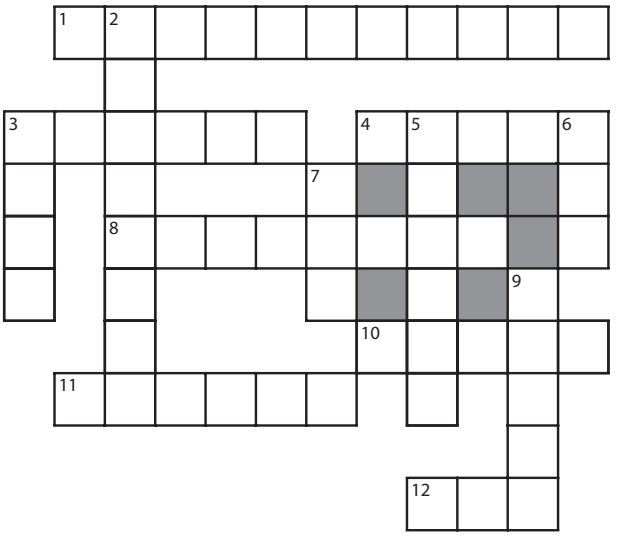
Achieve!

Genes and chromosomes

☞ **1** Complete the crossword puzzle.



Across

1. Objects inside the nucleus
3. The mother or father
4. Another word for 'dwarf'
8. A characteristic that overrides another
10. Hereditary factors affecting chromosomes
11. A cross that is not pure
12. Ova, sperm and pollen are examples of _____ cells

Down

2. Genetics is the scientific study of _____
3. Mendel's first experiments were with _____.
5. Scientist who made the first discoveries in genetics
6. Mendel's first _____ is how dominance works
7. deoxyribonucleic acid
9. DNA has a double _____ shape

2 Decide whether these statements are true or false.

Characteristics that hide for a generation are called dominant. _____

Genes are found on chromosomes. _____

Humans have 56 chromosomes. _____

How genes segregate is known as Mendel's first law. _____

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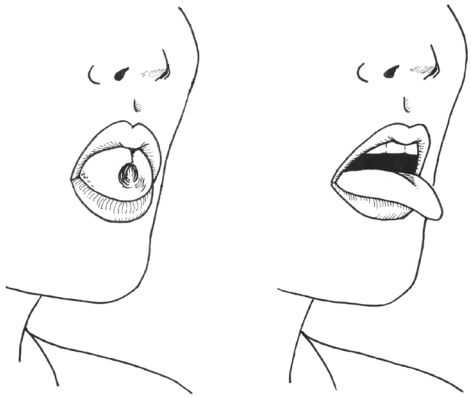
Roll your tongue

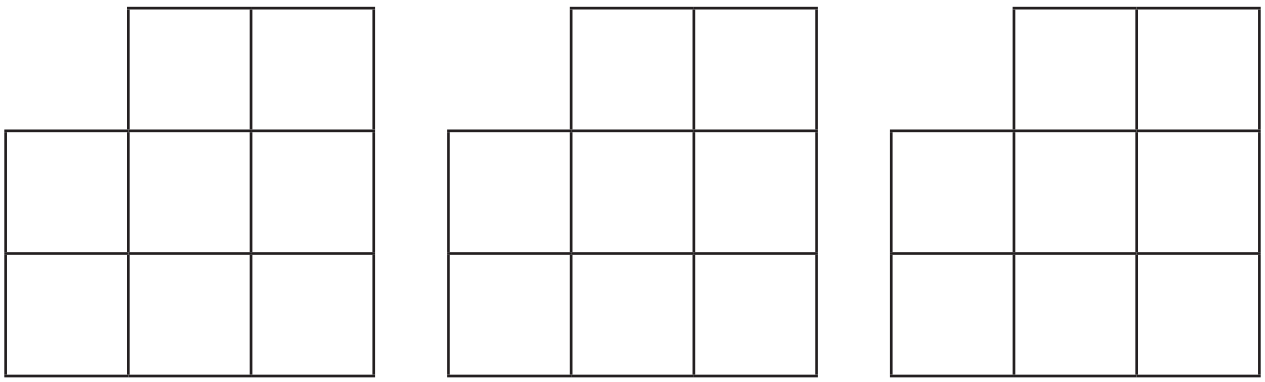


The ability to roll your tongue is **dominant** over the inability to roll your tongue.

☞ **1** If the genes for tongue rolling are written as **R** for the dominant (ability) gene and **r** for the **recessive** (inability) gene, write out the possible **genotypes** for tongue-rolling ability.

_____ _____

2 A man and woman can both roll their tongues. Show their possible genotype combinations and those of their children in the punnet squares below.

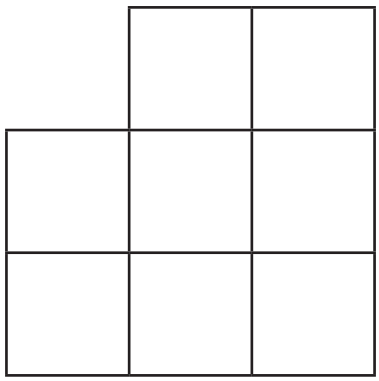


3 What are the **phenotypes** and the proportions of the children for each of the punnet squares? _____

4 Write out the possible genotypes if neither parent can roll their tongues.

_____

5 Fill out the punnet square below showing the children's genotypes.



6 What are the children's phenotypes? _____

_____

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Teacher's notes – Genetic changes

Achieve!

Objectives

- To understand that genes do sometimes change and this is called a mutation
- To be aware that some mutations are beneficial, some neutral and some harmful
- To know that mutations can lead to a species becoming resistant to specific chemicals
- To realise that interbreeding between species is possible, but the resulting hybrids are often infertile
- To understand that genetic engineering means removing genes from one species and inserting them into the genetic material of another species

Prior knowledge

Students should already be aware of the term 'mutation', but they may only associate it with freakish examples, such as two-headed snakes. They may also be aware of the term 'GM crops' or 'GM foods', but may not know what this means.

Background

Genes do sometimes change and this is called mutation. Not all mutations are harmful. Some can be beneficial and, in the case of mutations in pest species and microorganisms, can lead to a resistance to chemical controls and antibiotics. Scientists are able to manipulate genes and transfer them from one organism to another. This is called genetic modification or genetic engineering and is a controversial topic.

Starter activity

Ask students to consider the term 'mutation'. Gather the students' ideas about what this term means and list examples. Show them a picture of a harmless mutation, such as a 'rogue' or a 'sport' in a plant. This is where a branch of a shrub has flowers of a different colour to the rest of the plant. Explain that not all mutations are harmful.

Activity sheets

The Activity sheet 'Strange hybrids' explains that some foods, such as loganberries and tangelos, are actually hybrids. They are crosses between two different species. The term 'hybrid' can also be used to describe a cross between two cultivars or breeds. Students read a passage about hybrids and choose the correctly spelled word in each pair. There is also a table to fill out. As an additional exercise, students can list the main points in note form.

The 'Pesticide resistance' Activity sheet looks at the way insect pests, microorganisms and even plants can become resistant to specific chemicals if they are over-used. This page has a passage with missing vowels and a graph to analyse.

The Activity sheet 'Sickle cell anaemia' presents an example of a beneficial mutation. Students identify correctly spelled words in a passage about the sickle-cell trait and how being heterozygous for this allele can give resistance for malaria. Students are encouraged to work out a genetic cross and there are higher order questions about the topic.

'GM crops' looks at the use of genetically modified (or GM) crops, currently underway in many parts of the world, and the objections many people have to this practice. This Activity sheet outlines the perceived risks and benefits of GM crops. Students fill in missing vowels and match phrases on the topic. As an additional activity, students can expand the notes to write a passage on the topic.

Extension ideas

Ask students to think of a mutation that could be beneficial to their life. When sharing responses, ask if any of these beneficial mutations could cause problems.

Selected answers

Sickle cell anaemia

	Hb^A	Hb^S
Hb^A	Hb^{AA}	Hb^{AS}
Hb^A	Hb^{AA}	Hb^{AS}

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Strange hybrids

☞ **1** Circle the correctly spelled word in each pair.

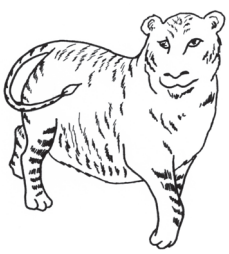
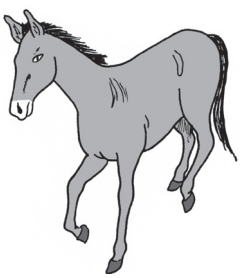
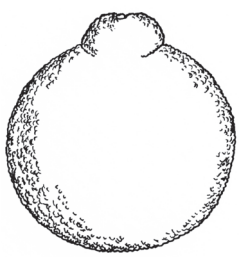
The word **hybrid** refers to the offspring/ofspring from a cross between two **genetically** different plants or animals. Most people use the word hybrid to lable/label the offspring of a domestic animal crossed with a wild animal, or a cross between spieces/species. For example, a wild duck crosed/crossed with a domestic duck, or the mule, which/witch is the offspring of a female horse and a male donkey. More **exotic** examples include the liger, which is the offspring of a lion and a tiger, and a zeedonk, produced by a zebra and a donkey. Many hybrids/hybreds are **sterile** and cannot reproduce. Plant hybrids are far more common/comon and often produce **viable** seed. Many corn varietes/varieties are hybrids. Other examples are loganberries, peppermint, tangelo/tanjello and grapefruit.

2 Use the word bank to fill in the table below.

Word bank

tiger liger tangerine donkey raspberry zeedonk

donkey horse mule blackberry tangelo

				
lion and t_____ cross	h_____ and d_____ cross	zebra and d_____ cross	b_____ and r_____ cross	pomelo and t_____ cross
l_____	m_____	z_____	loganberry	t_____

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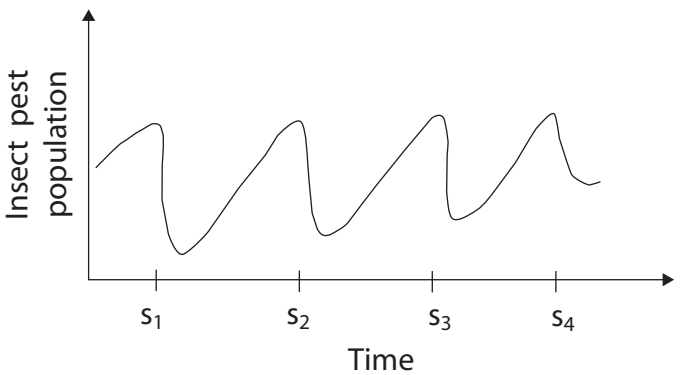
Pesticide resistance

Insect pests, microorganisms and even plants can become **resistant** to specific chemicals if they are over-used. Resistance can be caused by **mutations** in the genetic make-up of individuals.

☞ **1** Add the missing vowels (a, e, i, o, u) to complete the passage below.

Farmers spr__y their crops with **pesticides** to k__ll insect pests. Insects can quickly d__v__l__p resistance to p__st__c__d__s because they lay hundreds of eggs and they have such a sh__rt life span. In any given __ns__ct population a small proportion will have a **genetic** r__s__st__nc__ to a pesticide. After the p__st__c__d__ has been sprayed, m__st insects will die but the resistant ones will live to br__ __d with one another. The next g__n__r__t__ __ n will have a lot of resistant insects so when the f__rm__r sprays the same pesticide it will kill off f__w__r insects. Eventually, over r__p__ __t__d pesticide applications, the majority of the insect p__p__l__t__ __n will be resistant to the chemical.



* where s = spray application

2 Are all the pests killed after the first spray application? _____

_____

3 Why might some pests survive?_____

_____

4 Why do fewer insects die at each subsequent spray?_____

_____

5 How can the farmer better manage the insect population?_____

_____

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Sickle cell anaemia

☞ **1** Circle the correct word in each pair.

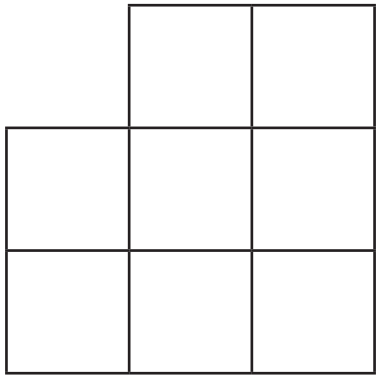
Many **genetic mutations** can be harmful to an organism/organizm but some can also be benificial/beneficial. In Africa a generic/genetic mutation known as sickle-cell trait has been found to provide resistance to malaria. Malaria is a serious disese/disease which kills millions of people/person every year. It is caused by a microscopic/microscope parasite which attacks read/red blood cells. People with sickle-cell trait have an increased resistance to milaria/malaria because their blood/bleed cells are oddly shaped and won't host the parasite/parasyte properly. Resistent/Resistant people are **heterozygous**, which means they carry only one **allele** of the sickle-cell trait. Those who are **homozygous** have two alleles/allelles of the sickle-cell trait. These individuals are weakened and have a short/shirt life-span. This condition is called sickle cell anaemia.

2 A sickle-celled red blood cell has a flattened shape and can't pick up **oxygen** as efficiently as a normal red blood cell. How would this create health problems for an individual?

_____

_____

3 The allele for sickle-cell trait is Hb^S and for regular blood cells it is Hb^A. Fill out the punnet square below showing a cross between a homozygous person (2 genes the same) and a heterozygous person (1 gene different to the other) for the sickle-cell allele.



4 What are the resulting **genotypes** and **phenotypes**? _____

_____

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GM crops

Genetically modified crops are food and fibre crops that have been modified by gene technology. External genes are inserted into a plant to change or improve its characteristics.

☞ **1** Add the missing vowels (a, e, i, o, u) to complete the notes below.

Potential benefits

- Improved **yields** and r__d__c__d damage from pests, d__s__ __s__s and weeds
- Reduced costs due to less p__st__c__d__ use
- More flexible f__rm__ng operations
- Less c__lt__v__t__ __n means better s__ __l **conservation**
- Reduction in pesticide p__ll__t__ __n

Potential risks

- P__ss__bl__ health effects
- New **allergens** could be cr__ __t__d
- Herbicide/pesticide r__s__st__nc__ may occur __v__r time
- Cross br__ __d__ng could occur with wild or non-GM cr__ps
- B__ __d__v__rs__ty affected as traditional f__ __d sources are lost
- Companies could c__ntr__l whole food crops
- Ethical/philosophical/religious __bj__ct__ __ns, i.e. scientists playing God

2 Match the following phrases.

GM insect-resistant cotton was	are against using GM crops.
Government agencies rigorously	introduced in Australia in 1996.
External genes are inserted	test GM crops for safety.
Organic food producers	into the crop plant.

3 Organic farmers have a special certificate declaring their food free of any additives, chemicals or artificial substances. Why would an organic farmer object to a neighbour growing GM crops? _____

_____

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Teacher's notes – Biodiversity

Objectives

- To understand the concept of biodiversity
- To know that habitats can range from those that are rich in species diversity to those that are poor
- To recognise that some countries have megadiversity status
- To become familiar with the concept of genetic diversity

Prior knowledge

Students should be aware that some habitats, such as tropical rainforests, support huge numbers of species. Plants, animals and microorganisms abound in their thousands. These habitats are 'biodiverse'.

Background

Biodiversity is a relatively new term that comes from the words 'biological' and 'diverse'. It is used to describe not only the diverse range of species in a given area but also the diverse range of genetic material within species. Megadiversity is a term coined to describe countries that have a vast range of species within their borders, many of which are endemic to those countries.

Starter activity

Working in groups, have students list the animals that live in a tropical rainforest. Students can name specific species, such as tigers, or they can write 'large carnivores'. Encourage them to come up with as many organisms as possible, including plants, insects and microorganisms. Ask them to compare their lists with the range of organisms that live in the schoolyard. Students could spend ten minutes outside naming as many plants, animals and insects as they can find.

Activity sheets

The Activity sheet 'Megadiverse Australia' introduces the concept of biodiversity, and explains that Australia is one of 17 megadiverse countries. This means that it contains a huge proportion of the world's biodiversity. Students are encouraged to think about animal and plant groups such as parrots and eucalypts. How many of these species does Australia have compared to other countries? Students look at five different Australian habitats and complete a table detailing these. Students may need prompting to complete the table. Base a class discussion around this towards the end of the session.

The 'Dasyurids vs kangaroos' Activity sheet introduces carnivorous Australian marsupials known as dasyurids. Dasyurids have evolved from genetic lines rich in diversity whereas kangaroos have evolved from a very narrow genetic base. Subsequently the extinction of a dasyurid species, such as the Tasmanian tiger, means that a huge amount of genetic information is lost to science. This work sheet has a passage for students to read and a table to fill out identifying various species as either kangaroos or dasyurids.

'Coral reefs vs salt flats' explains that some habitats support a huge range of organisms whereas others can be so harsh that species diversity is limited. This activity sheet presents two habitats and the species they support. Students should be able to fill out the table below the pictures, listing species diversity and drawing a comparison between the two habitats.

In the Activity sheet 'Australian freshwater turtles' students fill in the missing vowels in some notes about some of Australia's freshwater turtles. There are also some true/false statements to complete. As an additional activity students can expand the notes to write a paragraph on the topic.

Extension ideas

Ask students to research one of the other megadiverse countries of the world and find out how many species it supports.

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Achieve!

Megadiverse Australia

Biodiversity is a term used to describe the range of living things in an area. The word comes from the term 'biological **diversity**' and includes plants, animals and microorganisms.

☞ **1** Put the following organisms in the right category.

marsupials	parrots	eucalypts	protozoa
monotremes	fish	acacias	coral

Plants	Animals	Microorganisms

Some countries, especially those with **tropical** forests, such as Australia, have huge numbers of **unique** plants and animals. These countries are called **megadiverse**. Even though they cover a small amount of the Earth's surface, they support a large amount of its biodiversity. Australia also provides many different **climatic** conditions and **habitats**. This means that Australia is home to many different **ecosystems**, each with a diverse range of species.

☞ **2** Unique means **a)** native **b)** one of a kind **c)** special **d)** different.

3 Why are Brazil, the Congo and India megadiverse countries? _____

_____

4 The table below lists some habitats found in Australia. List as many Australian animals as you can for each habitat.

Coral reef	Desert	Tropical rainforest	Wooded grassland	Temperate rainforest
• shark • crab	• bilby • snake	• tree frog	• emu • echidna	• leech • lyrebird

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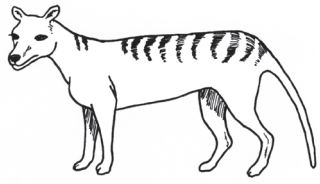
Dasyurids vs kangaroos

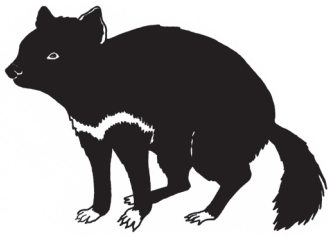
Genetic diversity is a term used to describe the variety of genes within any species. Some animals such as kangaroos are all genetically very similar. **Dasyurids**, or carnivorous marsupials, have evolved from ancient species and are more genetically diverse. Numbats, quolls and the Tasmanian devil are all dasyurids. If one species of dasyurid becomes **extinct**, we lose a large genetic resource. Several species of dasyurids are **endangered** and the Tasmanian Tiger is already extinct.

☞ **1** Label each of these species as either dasyurid or kangaroo.

wallaby _____

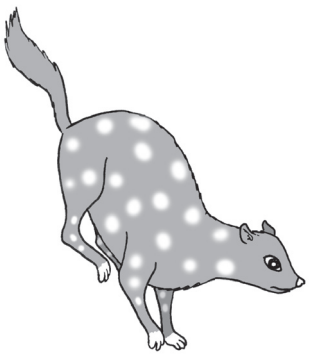


Tasmanian tiger _____

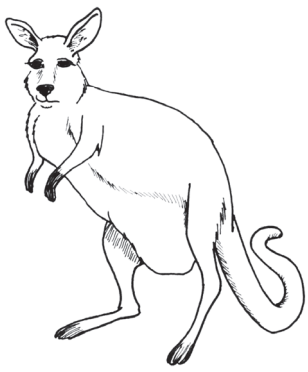


Tasmanian devil _____

red kangaroo _____



quoll _____



euro _____

numbat _____

grey kangaroo _____

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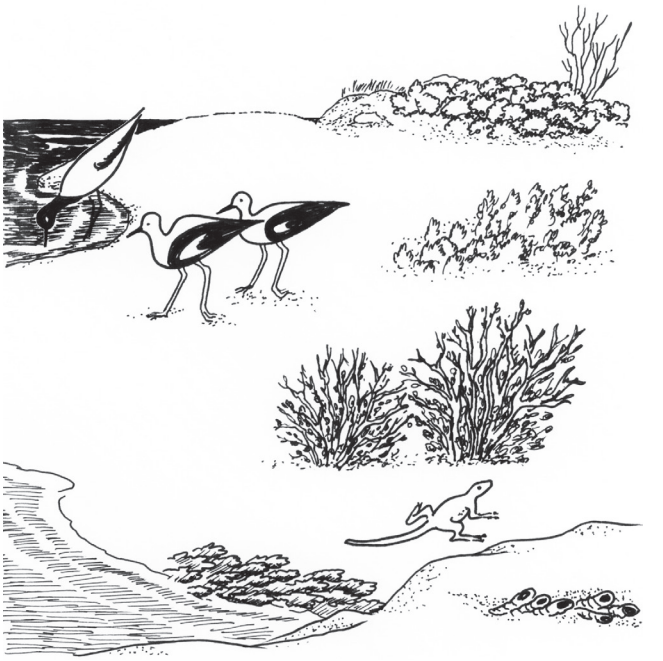
Coral reefs vs salt flats

Some **habitats** support a huge range of organisms, but others can be so harsh that only a few **species** can survive there.



☞ **1** Study the pictures above and list the different **flora** and **fauna** you can see.

Coral reef	Salt flat

2 Which habitat has more species diversity? _____

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Australian freshwater turtles

The Murray-Darling Basin is home to a **diverse** range of freshwater turtles.

☞ **1** Add the missing vowels (a, e, i, o, u) to the notes below.

The family Chelidae

- Twenty-four sp__c__ __s found in Australia
- Three d__ff__r__nt species in the Murray-Darling Basin alone

Broad-shelled t__rtl__

- Largest Australian freshwater turtle
- Found in slow-moving m__ddy water on river b__tt__ms
- Predator that __tt__cks fast moving prey: shrimp, sm__ll fish

Snake-necked turtle

- F__ __nd in swamps, b__ll__b__ngs and on land
- Eats __ns__ct larvae, larval **crustaceans**, d__ __d fish

Macquarie turtle

- Found in l__rg__ rivers, w__t__rh__l__s
- Eats __lg__ __, **aquatic** plants and insects, dead fish
- Basks on exposed l__gs or rocks

2 Decide whether each of these statements is true or false.

a) Freshwater turtles all eat the same things. _____

b) The Snake-necked turtle is a **carnivore**. _____

c) The Broad-shelled turtle is Australia's biggest freshwater turtle. _____

d) The Murray-Darling Basin has many different freshwater **ecosystems**. _____

e) The Macquarie turtle hides on murky river bottoms. _____

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Teacher's notes – Natural selection

Achieve!

Objectives

- To understand that individuals of a species show variation in characteristics
- To know that isolated individuals may change over time from the original population
- To understand that selective breeding can produce desired characteristics in domestic plants and animals
- To be aware that some animals select partners based on specific characteristics

Prior knowledge

Students should understand that there can be variation between individuals of a particular species. They should also be aware that purebred pets have particular desirable characteristics that animal breeders try to maintain in order to minimise variation.

Background

Individuals of a species show variation in characteristics. Some characteristics are better suited to a particular environment. For instance, light speckled moths are better camouflaged on mottled tree bark than on darkened trees. Individuals better suited to their environment are more likely to survive, reproduce and pass on their characteristics to the next generation. This is known as natural selection. Natural selection can result in a new species over time, especially if the population is isolated. Sexual selection is where animals choose a partner on the basis of a particular characteristic, such as mating call or courtship display. Selective breeding is when animal and plant breeders manipulate crosses in breeds or cultivars to produce a desired characteristic.

Starter activity

Show students pictures of various dog or cat breeds. Ask the students to list the characteristics of each breed and the ways they vary. Explain that the wolf was the ancestor of all modern dogs. How do Chihuahuas and Great Danes differ from wolves and from each other? Talk about variation in human characteristics. People on the First Fleet were generally shorter than Australians today, as evidenced by doorways and bunk sizes. How have we as a population grown taller?

Activity sheets

The Activity sheet 'Comparing apples with apples' demonstrates how individuals of a species can vary from one another. Different apple cultivars show wide variation within a species. Students are asked to identify similarities and differences. They are then asked to list differences between class members. As an additional activity students could draw a bar graph depicting the frequency of these differences. There is also a small cloze exercise.

The 'Isolated islands' Activity sheet examines the role that isolation plays in species evolution. Charles Darwin first studied finches in isolated islands and this comprehension activity outlines the basis of his work. Students fill in the missing vowels in the table before completing some true/false questions.

Individuals of a species can choose mates based on behavioural characteristics. This is best described by elaborate courtship rituals in different bird species. Australia has many species of bowerbird which build a small bower to attract females. The Activity sheet 'Sexual selection' presents cloze sentences on bowerbird courtship for students to complete. There are squashed sentences to separate out and a table to fill in. As an additional exercise students could expand the notes to write a paragraph on the topic. Additional questions on the topic could include 'What effect does sexual selection have on courtship behaviour in further generations?' and 'How does sexual selection contribute to the evolution of a species?'

Farmers have used selective breeding for centuries. The 'Selective breeding' Activity sheet presents students with information on how humans control breeding in animals and crops to maintain specific characteristics. Students match the phrases about selective plant breeding, then fill out a table listing desirable characteristics and genetic weaknesses in animal breeds.

Extension ideas

Students could each research a domestic animal with a genetic weakness and present their findings to the class.

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Comparing apples with apples

Variation describes the differences found in **individuals** of a species. Apples are all members of the same species: *Malus domestica*. However, they are not all the same. Look at the following apples and their **characteristics**.

Red Delicious	Pink Lady	Yellow Delicious	Granny Smith
• dark red skin • sweet flavour • five bumps on base • creamy white flesh	• pink and yellow skin • sweet flavour • crisp white flesh	• yellow skin • sweet juicy flavour	• green skin • tart flavour • large firm fruit

☞ **1** Which characteristics vary from one another? _____

_____

2 Which apples share a characteristic? _____

_____

3 Look at the differences between members of your class. Collect data detailing these differences in the table below. Add extra characteristics in the blank heading spaces.

Name	**Hair colour**	**Eye colour**	**Height**		

4 Do you think these differences are passed on from our parents, or are they due to how we live our lives? _____

_____

5 Complete the following sentences to understand the difference between **inherited** variation and **environmental** variation.

Word bank

variation characteristics live parents Differences inherited

D_____ in characteristics gained from our p_____ are called i_____ variation. Differences in c_____ due to the way we l_____ are called environmental v_____.

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Isolated islands

When Charles Darwin studied finch populations in the remote Galapagos Islands, he noticed that the **island** finches had different shaped beaks to the **mainland** finches from which they were descended. The mainland finches' beaks were suited to eating grain, which was scarce on the islands. The island finches' beaks were better suited to eating insects, fruit or leaves. Darwin believed that over many generations, island finches with diverse beak shapes survived better than those whose beaks were the same as the mainland finches.

Darwin proposed that the finches that had adapted so they could eat a variety of foods, rather than depending on the limited supplies of grain, survived and passed on their characteristics to their young. Because the finches were isolated, the genetic base was small and the beak characteristics of the mainland finches were bred out over time. Darwin described the survival of those species that changed their characteristics in order to survive as **natural selection**.

☞ **1** Scarce means **a)** plentiful **b)** scared **c)** very few **d)** damaged.

2 Add the missing vowels (a, e, i, o, u) to complete the table below.

	Beak shape	**Food type**
	r__ __nd__d b__ __k	b__ds __nd fr__ __t
	sh__rp __nd n__rr__w	__ns__cts
	sh__rp tr__ __ng__l__r	l__ __v__s

3 Some of these statements are correct, but some are untrue. Rewrite the false statements so they are true.

a) Animals' characteristics change over time.

_____

b) Isolated finches only eat grain.

_____

c) Charles Darwin studied finches in Australia.

_____

d) Isolated populations adapt to their environment.

_____

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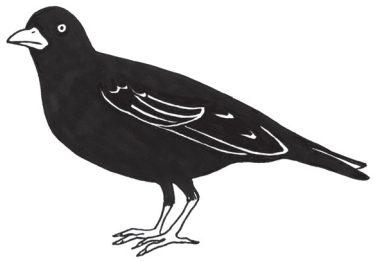
Sexual selection



Bowerbirds live in northern Australia and throughout Papua New Guinea. They are **polygamous**, which means they have more than one mate, and have unique **courtship** behaviour.

☞ **1** Use the word bank to complete these sentences about the courtship behaviour of bowerbirds.

Word bank							
sticks	best	attract	bower	coloured	many	steal	males

- Males build a structure called a b_____.
- They decorate the **bower** with s_____ and brightly c_____ objects.
- Bowers a_____ females.
- Females visit m_____ bowers.
- Males sometimes s_____ objects from one another's bowers.
- Females mate with m_____ that make the b____ bowers.

The male bowerbirds that make the most elaborate bowers are able to pass on their genetic material. This is called sexual selection, because the females select partners based on an ability, quality or strength.

☞ **2** Separate the words below to make sentences.

Malefrogsattractfemaleswiththeirloudcalls.

_____

Malelyrebirdsattractfemalesbydisplayingtheirlargetails.

_____

Malewildhorsesfightwitheachothertoleadtheherd.

_____

Theherdleaderistheonlymaletomatewithallthefemales.

_____

3 Select the most successful males in the following animal groups.

Aggressive stallion	Passive stallion
Quiet frog	Loud frog
Non-thieving bowerbird	Thieving bowerbird
Lyrebird with a small tail	Lyrebird with a big tail

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Selective breeding

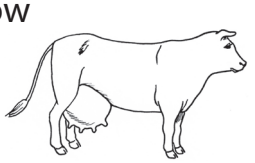
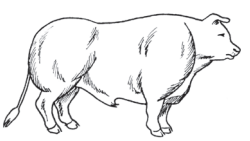
All dog breeds are **derived** from the same **genetic source** as the wolf. Over centuries, owners selectively chose dogs with specific **characteristics**, such as spots or long legs. Continuing to breed animals with particular characteristics meant that different types, or breeds, were established. This process is called **selective breeding**. It is highly controlled and leads to a small **genetic pool**. Because of the small gene pool, unwanted genetic diseases or weaknesses can occur.

☞ **1** Match the following phrases.

Selective breeding of the European	of fruit, flowers and vegetables.
Selective breeding reduces a	cabbage produced broccoli and cauliflower.
Plant breeders make new cultivars	species' gene pool.

2 Put the following characteristics and **complications** into the table.

weak hips huge heavy udders breathing difficulties
cute squashed faces big muscles for meat production can't walk comfortably
large calves and narrow birth canal graceful sloping back

	Desirable characteristics	**Complications**
Dairy cow 		
Beef cattle 		
Pug dog		
German shepherd dog		

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Teacher's notes – The fossil record

Objectives

- To recognise that plants and animals evolved over millions of years
- To know that scientists give names to periods of geological time
- To learn about an Australian dinosaur and prehistoric amphibian
- To recognise that algae is one of the oldest organisms on Earth

Prior knowledge

Students should know that dinosaurs existed between 200 million and 65 million years ago. Students may have knowledge of some prehistoric animals.

Background

The Earth is billions of years old. For much of this time there was little record of living things. These earliest times are called the Pre-cambrian age and represent more than 80% of geological time. Life evolved during the more recent eras: the Paleozoic (ancient), the Mesozoic, (middle) and the Cainozoic (modern). Throughout this vast amount of time organisms lived and died and were sometimes preserved as fossils in rocks. This is called the fossil record. Scientists can determine the age of various rocks depending on the types of fossils they contain.

Starter activity

Ask students what they know about dinosaurs. Were humans around in those days? What about birds, insects and flowers? Can the class make a timeline to place these organisms in geological time?

Activity sheets

The Activity sheet 'Eras and periods' introduces the different eras and periods of geological time. Students complete a table in order to familiarise themselves with these geological ages.

The 'Ancient stromatolites' Activity sheet looks at algae, which has existed on Earth for billions of years. One early organism found in the fossil record is the stromatolite, a small mound built by a type of algae. Stromatolites appear similar to a coral formation, but they are far older than coral. Students have a cloze passage to complete on this topic and some higher order questions to consider.

Many people are familiar with prehistoric animals such as dinosaurs and giant marsupials. Plants also evolved and diversified over time. The Activity sheet 'Prehistoric plants' presents students with a table showing the evolution of plants throughout history for them to complete. There is also a small passage to take notes from as an additional activity.

In the 'Giant amphibians' Activity sheet, students form a paragraph by expanding notes on *Koolasuchus cleelandi*, an Australian prehistoric amphibian. Its fossils were found on the Victorian coast in recent years. Students also draw a timeline of the Mesozoic era and show where various organisms first evolved.

The Activity sheet '*Muttaburrasaurus langdoni*' introduces a plant-eating dinosaur that once roamed Australia. Students complete a cloze passage about this dinosaur as well as answering true/false statements and matching some phrases.

Extension ideas

Give each student a nametag with the name of a particular organism from the fossil record, such as conifers, algae, cycads, Muttaburrasaurus, Diplodocus, woolly mammoth. Ask students to arrange themselves from earliest to latest in a 'student' geological timeline.

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Eras and periods

Scientists name chunks of geological time according to three broad **eras**. These are modern (**Cainozoic**), middle (**Mesozoic**) and ancient (**Paleozoic**). Each of these eras is divided into **periods**.

☞ **1** Use the word bank to complete the table below.

Word bank

Jurassic	insects	middle	species	fish
mammals	Devonian	Extinction	birds	Tertiary

Era	Period		Description
Cainozoic (recent life) modern to 65 million years ago	Quaternary		• Mostly modern s_____ • **Megafauna** around the world
	T_____		• Modern animal groups • Australia separates from Antarctica • Expansion of m_____ throughout the world
Mesozoic (middle life) 65 million to 250 million years ago	Cretaceous	late	E_____ of many groups of vertebrates
		m_____	• World's earliest known marsupials • Australian giant amphibian *Koolasuchus cleelandi*
		early	Australian dinosaur *Muttaburrasaurus langdoni*
	J_____		World's oldest b_____
	Triassic		World's oldest mammals
Palaeozoic 250 million to 542 million years ago	Permian		**Glaciers** throughout Australia
	Carboniferous		• World's oldest reptiles • Giant i_____
	D_____		World's oldest amphibians
	Silurian		Land turns green
	Ordovician		World's first f_____
	Cambrian		World's oldest vertebrates

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Ancient stromatolites

☞ **1** Use the word bank to complete the passage below.

Word bank					
fossil	extinct	algae	seen	period	stromatolites
ago	limit	salt	beds	populated	layered
world	million	marine	water		

Most organisms that p_____ the Earth millions of years ago are now e_____. Modern plants and animals have only been around for the last few m_____ years.

One organism that was common throughout the world in the **Precambrian** p_____, more than 570 million years a____, was the **stromatolite**. Stromatolites are thought to be a major part of the f_____ record for the first 3.5 billion years of life on Earth. Stromatolites are l_____ growing structures formed in shallow s_____ water. Microorganisms such as **cyanobacteria**, also known as blue-green a_____, form these growths by binding and cementing grains of sand to themselves. Over time the **microorganisms** build little mounds or b_____ in the water.

Most s_____ have vanished from the Earth and can only be s____ as fossils. However, Western Australia's Hamelin Pool is one of only three sites in the w_____ that still has living m_____ stromatolites. The w_____ is twice as salty as normal sea water so few predators or competitors l_____ the microbes' growth.

2 How are stromatolites different from rocks? _____

_____

3 Why is Hamelin Pool a safe environment for stromatolites? _____

_____

4 How could rising sea levels affect the survival of stromatolites? _____

_____

_____

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Prehistoric plants

Word bank

mosses
Devonian
flowering
Jurassic
grassland
conifers
Permian

The earliest plants on Earth were marine algae that lived billions of years ago. During the Silurian period plants appeared on land. They were leafless **vascular** plants called psilophytes. Early mosses also appeared around the same time. The next plants to **evolve** were ferns and conifers, cycads and gingkoes. Many dinosaurs ate these species. Flowering plants and grasses are relatively modern in **geological** time. They developed during the Cretaceous period.

☞ **1** Use the word bank to fill in the table.

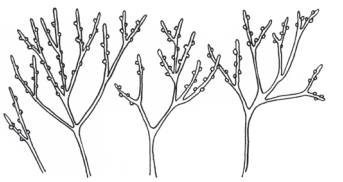
Time period	Plant forms
Tertiary	g_____ expands, forests shrink
Cretaceous	early f_____ plants, modern trees
J_____	cycads, conifers and gingkoes
Triassic	ferns, cycads, gingkoes and c_____
P_____	ferns and conifers
Carboniferous	club m_____, horsetails, ferns and cordaites
D_____	lowland forests of giant psilophyta plants

2 Match the pictures with the following statements.

Psilophyta plants were common in the Devonian period.

Grasses grow today all over the world.

Flowering plants are very recent types of plants.



Ferns are a very ancient type of plant.

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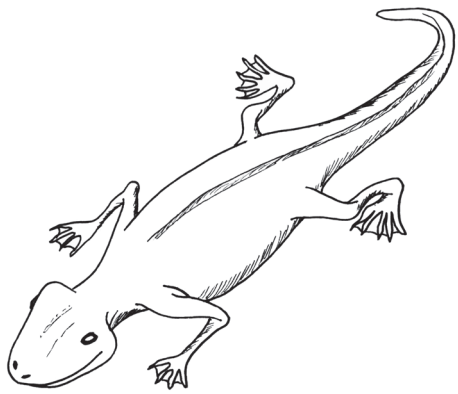
Giant amphibians

The **Mesozoic era** was the age of dinosaurs. Dinosaurs roamed Australia then, as well as some strange **amphibians**.

☞ **1** Use the phrases below to write a paragraph about a fossil of a giant prehistoric amphibian from the Mesozoic era that was found in Australia.

- jawbone discovered on a Victorian beach in 1990 by scientist Mike Cleeland
- lived in the swamps of prehistoric Australia
- an amphibian the size of a saltwater crocodile
- scientific name: *Koolasuchus cleelandi*
- ate fish and shellfish
- had a large, flat, broad head and a long body
- from the middle Cretaceous period, about 110–100 million years ago



The Mesozoic era is divided into three **periods**. The earliest is the **Triassic period** (230–195 million years ago), when the first mammals appeared. The next is called the **Jurassic period** (195–141 million years ago), when the world's first birds appeared. The last is the **Cretaceous period**, which ended around 65 million years ago.

☞ **2** Draw a timeline showing the three periods of the Mesozoic era. Show when birds, mammals and *Koolasuchus cleelandi* lived.

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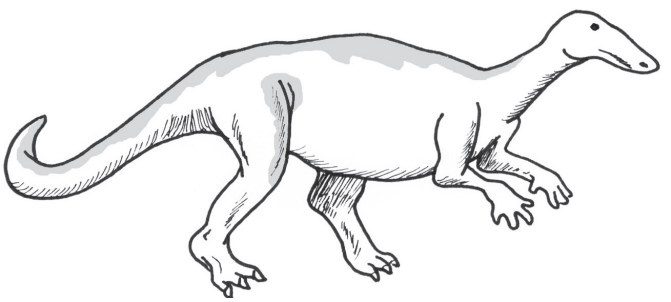
Muttaburrasaurus langdoni

☞ **1** Use the word bank below to complete the passage.

Word bank					
Scientists	dinosaur	seven	snout	rare	could
skeleton	ate	central	inflated	Cretaceous	

Complete dinosaur skeletons are very r_____. Most d_____ **skeletons** found in Australia are incomplete. *Muttaburrasaurus langdoni* was found in c_____ Queensland in 1963. It consisted of a **skull** and most of the s_____, minus the tail. *Muttaburrasaurus* lived during the C_____ period in northern Australia. Scientists believe it was about s_____ metres long. It walked on its two hind legs but c_____ browse on all fours. Its distinctive feature was an i_____ bony roof over the s_____ in front of the eyes. S_____ believe that it a____ plants, as its teeth were suitable for shearing and chewing.



2 Write whether these statements are true or false.

a) *Muttaburrasaurus langdoni* was a rare dinosaur. _____

b) *Muttaburrasaurus* probably ate plants and animals. _____

c) *Muttaburrasaurus* was a horned dinosaur. _____

d) Complete dinosaur skeletons are hard to find. _____

e) *Muttaburrasaurus* could walk on four legs or two. _____

3 Match the phrases below.

Muttaburrasaurus was discovered	the Cretaceous period.
Muttaburrasaurus probably	near Muttaburra in Queensland.
Muttaburrasaurus lived in	strangely shaped nose.
Muttaburrasaurus had a	travelled in large herds.

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